Reinforcement Learning for Continuous-Time Mean-Variance Portfolio Selection in a Regime-Switching Market

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Abstract

We propose a reinforcement learning (RL) approach to solve the continuous-time mean-variance portfolio selection problem in a regime-switching market, where the market regime is unobservable. To encourage exploration for learning, we formulate an exploratory stochastic control problem with an entropy-regularized mean-variance objective. We obtain semi-analytical representations of the optimal value function and optimal policy, which involve unknown solutions to two linear parabolic partial differential equations (PDEs). We utilize these representations to parametrize the value function and policy for learning with the unknown solutions to the PDEs approximated based on polynomials. We develop an actor-critic RL algorithm to learn the optimal policy through interactions with the market environment. The algorithm carries out filtering to obtain the belief probability of the market regime and performs policy evaluation and policy gradient updates alternately. Empirical results demonstrate the advantages of our RL algorithm in relatively long-term investment problems over the classical control approach and an RL algorithm developed for the continuous-time mean-variance problem without considering regime switches.

Keywords: reinforcement learning, actor-critic, mean-variance, portfolio selection, partial information, regime-switching, Wonham's filter

1 Introduction

Initiated by the seminal work of Markowitz (1952) for portfolio selection in a single period, mean-variance (MV) is an important criterion for portfolio choice. This criterion captures the tradeoff between expected return and risk in a simple and intuitive way, making it popular in practice. Ever since Markowitz (1952), the MV investment problem has been further studied in the discrete-time multi-period setting (Li and Ng (2000)) and the continuous-time setting (Zhou and Li (2000)), along with many other generalizations. A comprehensive survey of this extensive literature can be found in Cui et al. (2022).

The study of portfolio selection problems in the literature typically takes the classical stochastic control approach, where an optimal control problem is formulated under a given stochastic model of asset prices and solved to obtain the optimal investment policy. To implement this approach in practice, one first estimates the model parameters and then plugs them into the optimal control function. Due to the estimation error, the investment policy obtained is no longer optimal even when there is no model misspecification. Furthermore, it is well-known that any error from model estimation is prone to be amplified in the optimization procedure, which can make the "optimal" policy from the classical approach far from optimal in practice.

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In this paper, we attempt to solve the MV portfolio selection problem by reinforcement learning (RL), which has become increasingly popular for financial applications; see Hambly et al. (2022) and Jaimungal (2022) for surveys of RL in finance. Unlike the classical stochastic control approach where the optimal control is fixed by estimated model parameters from historical data, RL adopts a trial-and-error approach, which tries to improve the policy adaptively through interactions with the environment. Thus, it provides a promising way to overcome the drawback of the classical approach.

Although the majority of RL research focuses on the discrete-time setting (see e.g., Sutton and Barto (2018)), there is growing interest in continuous-time RL. The continuous-time setting is natural for problems with decisions made in high frequency. Furthermore, it allows one to obtain useful representations of the optimal value function and optimal policy in some problems, which can be exploited to facilitate learning. These representations obviate the need to use a general function such as a neural network to approximate the value function and policy, which often requires a large number of parameters and can make learning problematic.

To do RL in continuous time and space under a general diffusion setting, Wang et al. (2020) proposed an exploratory stochastic control framework with entropy regularization, where the control is allowed to be stochastic. An impressive general theory was then developed in Jia and Zhou (2022a,b,c) for policy evaluation (PE), policy gradient (PG) calculation for actor-critic learning, and q-learning (the continuous-time counterpart of standard discrete-time Q-learning). Wang and Zhou (2020) proposed an RL algorithm known as EMV to learn the optimal precommitted policy for the continuous-time MV portfolio selection problem under the geometric Brownian motion (GBM) model for the risky asset price. They obtained analytical representations of the optimal value function and optimal policy of the entropy-regularized MV problem, which provided them with natural parametrizations for learning. The EMV algorithm was later improved in Jia and Zhou (2022b) by using the martingale approach for learning and an extensive empirical study was carried out in Huang et al. (2022) by comparing it with deep learning based RL algorithms. It is well-known that the precommitted strategy of the MV problem is time inconsistent. To obtain a time-consistent strategy, Dai et al. (2020) developed an RL algorithm under a factor-based diffusion price model with observable factors to learn the equilibrium MV strategy based on a semi-analytical representation. Continuous-time RL algorithms have also been developed for other control problems; see e.g., Guo et al. (2022) and Firoozi and Jaimungal (2022) for mean-field games, and Wang et al. (2023) for optimal execution.

In this paper, we aim to learn the optimal precommitted strategy of the continuous-time MV problem in a regime-switching market. It has been well documented that the state of the financial market can change over time and such regime-switching behavior is too important to ignore in relatively long-term investment horizons; see e.g., Guidolin and Timmermann (2007, 2008), Tu (2010) and Ang and Timmermann (2012). Zhou and Yin (2003) and Yin and Zhou (2004) investigated the MV problem with regime switching in both discrete-time and continuous-time models and obtained analytical results for the efficient frontier as well as economic implications. The market regime is assumed to be observable in these papers. However, investors cannot directly observe the market regime in reality. What they can do is to infer its state from the observed information, which is a filtering problem. Dealing with the unobservable regime case brings significant challenges, as the investor only has partial information to make decisions. Xiong and Zhou (2007) studied the continuous-time MV problem with partial information and established the separation principle, which allows the investor to separate the problem into filtering and optimization. Elliott et al. (2010) developed an expectation-maximization (EM) algorithm for filtering in the regime-switching MV problem with unobservable regimes. Studies for investment problems with other objectives in the unobservable regime case include Sass and Haussmann (2004) and Rieder and Bäuerle (2005) for utility maximization and Dai et al. (2010, 2016) for trend following trading.

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We consider the realistic situation where the market regime is unobservable. The separation principle and the general theory of continuous-time RL established in Jia and Zhou (2022b) provide the foundations for developing an RL algorithm in our problem. We first try to develop some representations that would be useful for designing RL algorithms. To this end, we consider the exploratory stochastic control framework introduced by Wang et al. (2020). We obtain semi-analytical representations of the optimal value function and optimal policy, which involve two unknown functions as solutions to two linear parabolic partial differential equations (PDEs). We establish the properties of these two functions by using tools from the PDE theory and propose to approximate them based on polynomials. The representation of the optimal policy shows that it is Gaussian and gives us insights into its properties. It also suggests that we can focus on Gaussian policies for learning. We utilize these representations together with polynomial-based approximations to the PDE solutions to parametrize the value function and policy and construct an actor-critic algorithm for learning. In every interaction with the environment over the investment horizon, our algorithm first carries out filtering to obtain the belief probability of the market regime, which serves as a state variable for investment. It then conducts PE and PG updates alternately to learn the optimal policy. The separation of the filtering and learning steps is valid as the separation principle holds for the MV problem.

Our work can be viewed as an extension of Wang and Zhou (2020) to the regime-switching case, which is, however, nontrivial. They considered the GBM model for the risky asset, under which the exploratory MV problem can be solved analytically and the properties of the solution can be easily derived from the analytical formula. In comparison, after introducing regime switching, the exploratory MV problem admits no analytical solution for its optimal value function and optimal policy. In this case, properties of the solution cannot be easily obtained and must be derived using further mathematical analysis. We also demonstrate that it is practically meaningful to introduce regime switching for learning the MV policy. In an empirical study based on real market data, our algorithm outperforms the EMV algorithm by substantially reducing the out-of-sample variance in investment. The superior performance of our algorithm can be explained by its ability to utilize the guess of the market regime to improve investment performance, whereas the policy learned by EMV is completely ignorant of the state of the market. As a result, investment under the policy learned by our algorithm can grow faster in the bull market and suffer smaller losses in the bear market.

Our research is closely related to an extensive literature on RL with partial information, which focuses on the study of partially observable Markov decision processes (POMDPs) in discrete time; see e.g., the book by Krishnamurthy (2016) and the references therein. Several papers developed RL algorithms for investment problems with partial information in the discrete-time setting. Asiain et al. (2018) and Liu et al. (2020) formulated the investment problem as a POMDP, but the partial information issue is not caused by regime switching. Maringer and Ramtohul (2012) and Maringer and Zhang (2014) proposed a regime-switching recurrent RL model for investment, but the regime is assumed to be observable. To the best of our knowledge, the study of RL for investment problems in a regime-switching environment with unobservable regimes is still lacking, and our paper fills this gap in the continuous-time setting for the MV objective.

The rest of the paper is organized as follows. Section 2 presents the MV portfolio selection problem with regime switches and introduces an exploratory version of the problem. Section 3 derives the optimal value function and optimal policy of the exploratory problem. Section 4 develops an actor-critic RL algorithm based on the semi-analytical representations in Section 3. Section 5 and Section 6 conduct simulation and empirical studies, respectively. The last section concludes. The appendix provides an extension, a numerical scheme, and proofs.

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2 MV Portfolio Selection with Regime Switches

We consider the continuous-time MV portfolio selection problem over a fixed investment horizon $[0, T]$, which solves the following constrained optimization problem:

$$\begin{cases} \text{minimize} & \mathrm{Var}\left[X_T^u\right], \\ \text{subject to} & \mathbb{E}\left[X_T^u\right] = z, \\ & u := \{u_t : 0 \le t \le T\} \text{ is self-financed and admissible,} \end{cases} \tag{1}$$

where $\{X_t^u, 0 \le t \le T\}$ is the discounted wealth process under investment strategy u and its dynamics will be specified below, and z is the investment target set at $t = 0$ for the expected discounted terminal wealth. It is well-known that the problem is time inconsistent due to the variance term in the objective, and there is generally no dynamically optimal solution. In this paper, we focus on finding the optimal precommitted strategy, which is optimal only at $t = 0$.

Following Zhou and Li (2000), we introduce a Lagrange multiplier to transform the constrained problem into an unconstrained problem as

$$\min_u \mathbb{E}\left[(X_T^u)^2\right] - z^2 - 2\omega\left(\mathbb{E}\left[X_T^u\right] - z\right) = \min_\pi \mathbb{E}\left[(X_T^u - \omega)^2\right] - (\omega - z)^2. \tag{2}$$

The optimal solution to this unconstrained problem denoted by $u^* = \{u_t^*, 0 \le t \le T\}$ depends on ω. The optimal multiplier ω^* can be obtained by solving $\mathbb{E}\left[X_T^{u^*}\right] = z$. Our focus is on how to solve the uncontrained problem (2) for a fixed ω by RL in a regime-switching market.

2.1 Market Environment

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For simplicity, we consider a financial market with one risk-free asset and one risky asset for now. We deal with the multi-asset case in Appendix A. We assume that the interest rate paid by the risk-free asset is given by a constant $r \ge 0$, and the risky asset price process $S := \{S_t : t \ge 0\}$ is a regime-switching geometric Brownian motion (RS-GBM), which follows the stochastic differential equation (SDE)

$$dS_t = S_t\left(\mu_t dt + \sigma dB_t\right), \tag{3}$$

where $B := \{B_t : t \ge 0\}$ is a one-dimensional standard Brownian motion, the volatility $\sigma > 0$ is constant, and the drift μ_t is determinted by the market regime I_t. We model the market regime process $I := \{I_t : t \ge 0\}$ by a continuous-time Markov chain (CTMC) with state space $\{1, 2\}$, which is independent of the Brownian motion B. The two-regime case is often considered in the literature and in practice, which interprets one regime as a bull market and the other as a bear market.

For the CTMC, we denote its generator matrix by Q, which is given by

$$Q = \begin{pmatrix} -\lambda_1 & \lambda_1 \\ \lambda_2 & -\lambda_2 \end{pmatrix}, \tag{4}$$

where $\lambda_i > 0$ $(i = 1, 2)$ stands for the transition rate from regime i to the other. We assume the drift is constant in each regime and its value is given by μ_i in regime i. Define

$$\rho_i := (\mu_i - r)/\sigma, \tag{5}$$

which is the Sharpe ratio of the risky asset in regime i. In this paper, we call σ, μ_i, λ_i $(i = 1, 2)$ as environment parameters.

While we can observe the price of the risky asset at any time, the market regime is unobservable in practice. Let $\mathcal{F}_t^S$ be the sigma-field generated by the price process S up to time t. The regime process I and the standard Brownian motion B are not adapted to the filtration

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$\{\mathcal{F}_t^S : t \geq 0\}$. This makes the MV problem more challenging because the investor only has partial information about the states that matter for investment decisions.

Let u_t be the discounted dollar value invested in the risky asset at time t. The process $u = \{u_t : t \in [0, T]\}$ is an investment strategy, which is adapted to $\{\mathcal{F}_t^S : t \in [0, T]\}$ and satisfies the self-financing assumption and other standard conditions for admissibility. The discounted wealth process $X^u = \{X_t^u : t \in [0, T]\}$ under strategy u satisfies

$$dX_t^u = \sigma u_t \left(\rho_t dt + dB_t\right), \ t \in [0, T], \tag{6}$$

where the Sharpe ratio ρ_t depends on the market regime and takes value from $\{\rho_1, \rho_2\}$.

Remark 1. In the study of continuous-time regime-switching models, it is quite standard to let the volatility process independent of the regime because it can be exactly determined by the quadratic variation process of $\{\ln S_t : t \geq 0\}$ (Xiong and Zhou (2007)). In our framework, we assume σ is a constant to obtain tractability for the stochastic control problem. Regime-switching models with unobservable regimes and constant volatility are also used in Sass and Haussmann (2004) and Rieder and Bäuerle (2005) for utility maximization, Dai et al. (2010, 2016) for the optimal trend following rule, and Elliott et al. (2010) for the MV problem.

2.2 Filtering and Separation Principle

A general idea of dealing with unobservable states is to filter their values from the observed information. In our problem, we consider

$$p_t := P(I_t = 1 \mid \mathcal{F}_t^S),$$

which is the probability of the market being in regime 1 given the price information up to time t. It is also called the belief state in the POMDP literature and we will use this term below.

Consider the innovation process defined by

$$d\hat{B}_t := \frac{d\ln\left(S_t\right) - \left(\hat{\mu}_t - \sigma^2/2\right) dt}{\sigma}, \tag{7}$$

where

$$\hat{\mu}_t := \left(\mu_1 - \mu_2\right) p_t + \mu_2$$

is the filtered drift at time t. We also consider the filtered Sharpe ratio

$$\hat{\rho}_t := \frac{\hat{\mu}_t - r}{\sigma} = \left(\rho_1 - \rho_2\right) p_t + \rho_2. \tag{8}$$

It can be proved that $\hat{B} = \{\hat{B}_t : t \geq 0\}$ is a one-dimensional standard Brownian motion with respect to $\mathcal{F}^S$; see e.g., Kallianpur (2013), Theorem 8.1.2.

It is important to note that (6) shows the dynamics of the discounted wealth under the full information case in which the regime can be observed. However, due to partial information, it cannot be used for further analysis. Using (7) and (8), we can obtain the dynamics of the discounted wealth adapted to the observable filtration $\mathcal{F}^S$ as

$$dX_t^u = \sigma u_t \left(\hat{\rho}_t dt + d\hat{B}_t\right), \ t \in [0, T]. \tag{9}$$

With (9), the MV problem can be solved as in the complete information case. In the literature, this result is called the separation principle, which separates the stochastic control problem under partial information into filtering and control under complete information. The validity of the separation principle for the MV problem with partial information was established in Xiong and Zhou (2007) for general price dynamics with unobservable drift, which is not necessarily modulated by a CTMC.

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The dynamics of the belief state is given by the Wonham filter (Wonham (1964)) as

$$dp_t = \left[-\left(\lambda_1 + \lambda_2\right) p_t + \lambda_2\right] dt + \left(\rho_1 - \rho_2\right) p_t \left(1 - p_t\right) d\hat{B}_t, \tag{10}$$

which is adapted to the observable filtration $\mathcal{F}^S$. The belief state process $\{p_t : t \geq 0\}$ is a diffusion process with two boundary points given by 0 and 1. Using the classification of boundaries (Karlin and Taylor (1981)), it is shown in Dai et al. (2010) that both 0 and 1 are entrance boundaries. This means that when the process starts from 0 or 1, it enters the open interval $(0, 1)$. However, once the process is in $(0, 1)$, it has zero probability of hitting 0 and 1 in finite time. We assume the starting value $p_0 \in (0, 1)$ in our analysis, which implies that the state space of the belief state process is the open interval $(0, 1)$, not including the two endpoints. The drift in the SDE (10) for p_t can be rewritten as

$$\left(\lambda_1 + \lambda_2\right) \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} - p_t\right),$$

which shows that p_t reverts to the level given by $\lambda_2/(\lambda_1 + \lambda_2)$, the stationary probability of the market being in regime 1. Furthermore, the speed of mean-reversion is given by the sum of the two transition rates of the CTMC.

Xiong and Zhou (2007) developed a general theory of the MV problem in partial information with general unobserved stock drift dynamics. A particle system representation was established for evaluating the filter and a backward SDE was numerically solved to reach the optimal target wealth. Traditional approaches to address this problem involve filtering the unknown values of μ_i and λ_i, i.e., estimating their values based on the observable filtration; see e.g., Elliott et al. (2008) for filtering algorithms for hidden Markov models.

In this paper, we develop an RL approach to learn the optimal precommitted strategy under partial information from interactions with the market environment.

2.3 Regime-Switching Exploratory MV Problem

A basic feature of RL is exploration. To consider exploration, Wang et al. (2020) introduced an exploratory stochastic control framework in continuous time. The control u_t at time t is now drawn from a probability distribution over the control space. That is, control actions are randomized according to a stochastic policy. Specifically, we consider stochastic policies of the feedback form: $\boldsymbol{\pi} : (t, x, p) \in [0, T] \times \mathbb{R} \times (0, 1) \mapsto \boldsymbol{\pi}\left(\cdot \mid t, x, p\right) \in \mathcal{P}(\mathbb{R})$, where $\mathcal{P}(\mathbb{R})$ denotes the class of probability density functions on $\mathbb{R}$. It should be noted that in this paper, the bold letter $\boldsymbol{\pi}$ refers to a stochastic policy while π stands for the mathematical constant Pi. For any feedback distribution $\boldsymbol{\pi}$, we consider its differential entropy given by

$$\mathcal{H}(\boldsymbol{\pi}\left(\cdot \mid t, x, p\right)) = -\int_{\mathbb{R}} \boldsymbol{\pi}(u \mid t, x, p) \ln \boldsymbol{\pi}(u \mid t, x, p) du.$$

We assume that action randomization is independent of $\hat{B}$. Jia and Zhou (2022b) provide a mathematical framework to redefine the filtered probability space by introducing information from action randomization and we apply it to our problem. For any $t \in [0, T]$, let $\overline{\mathcal{F}}_t = \mathcal{F}_t^S \vee \sigma\left(Z_s, 0 \leqslant s \leqslant t\right)$, where $\{Z_s, 0 \leqslant s \leqslant t\}$ is a process of mutually independent random variables uniformly distributed over $[0, 1]$ (see Sun (2006)). We define $\overline{\mathbb{P}}$ as the probability measure on $\mathcal{F}_T$ extended from $\mathbb{P}$, and the two probability measures coincide when restricted to $\mathcal{F}_T^S$. The original filtered probability space $\left(\Omega, \mathcal{F}, \mathbb{P}; \{\mathcal{F}_t^S : t \in [0, T]\}\right)$ is now expanded to $\left(\Omega, \mathcal{F}, \overline{\mathbb{P}}; \{\overline{\mathcal{F}}_t : t \in [0, T]\}\right)$.

The discounted wealth process under stochastic policy $\boldsymbol{\pi}$ is now given by

$$dX_t^{\boldsymbol{\pi}} = \sigma u_t^{\boldsymbol{\pi}} \left(\hat{\rho}_t dt + d\hat{B}_t\right), \ t \in [0, T],$$

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where $u_t^{\boldsymbol{\pi}}$ is sampled from $\boldsymbol{\pi}(\cdot \mid t, X_t^{\boldsymbol{\pi}}, p_t)$. To encourage exploration, we introduce entropy regularization and solve the following regime-switching exploratory MV problem (hereafter, RSEMV problem) for a fixed ω:

$$\min_{\boldsymbol{\pi}\in\Pi} \overline{\mathbb{E}}\left[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda \int_0^T \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid t, X_t^{\boldsymbol{\pi}}, p_t)\right) dt\right] - (\omega - z)^2, \tag{11}$$

where Π is the set of admissible feedback control distributions to be specified below, and $\overline{\mathbb{E}}$ denotes expectation under $\overline{\mathbb{P}}$, which averages out both the noise of the environment as represented by $\hat{B}$ and the randomness in the actions. The weight $\Lambda \geq 0$ for the entropy regularization represents the willingness to explore and is also known as the temperature parameter in the RL literature. We call $X_t^{\boldsymbol{\pi}}$ the sample state at time t.

For theoretical analysis and deriving solutions to control problems under stochastic policies, Wang et al. (2020) introduced the exploratory state process, whose trajectories can be viewed as the average of the trajectories of the sample state process over infinitely many randomized actions. We now derive the dynamics of the exploratory state process, $\{\widetilde{X}_t^{\boldsymbol{\pi}} : t \in [0,T]\}$, in our problem. Consider $\Delta X_t^{\boldsymbol{\pi}} := X_{t+\Delta t}^{\boldsymbol{\pi}} - X_t^{\boldsymbol{\pi}}$ and $\Delta p_t := p_{t+\Delta t} - p_t$. Using the law of large number argument in Wang et al. (2020), we obtain

$$\begin{aligned}
\overline{\mathbb{E}}\left[\Delta X_t^{\boldsymbol{\pi}}\right] &= \int_{\mathbb{R}} \sigma\hat{\rho} u \boldsymbol{\pi}(u) du \Delta t, \\
\overline{\mathbb{E}}\left[(\Delta X_t^{\boldsymbol{\pi}})^2\right] &= \int_{\mathbb{R}} \sigma^2 u^2 \boldsymbol{\pi}(u) du \Delta t + o(\Delta t), \\
\overline{\mathrm{Cov}}\left(\Delta X_t^{\boldsymbol{\pi}}, \Delta p_t\right) &= \int_{\mathbb{R}} (\rho_1 - \rho_2) p(1-p) \sigma u \boldsymbol{\pi}(u) du \Delta t + o(\Delta t),
\end{aligned}$$

where $\lim_{\Delta t \to 0} o(\Delta t)/\Delta t = 0$. Note that in the above expectations, the randomness in the actions has been averaged out. It is also worth mentioning that unlike the MV problem without regime-switching studied in Wang and Zhou (2020), we must also consider the covariance between $\Delta X_t^{\boldsymbol{\pi}}$ and Δp_t in our problem.

As diffusion processes are essentially determined by the first two moments, we obtain the following dynamics for the exploratory state:

$$\begin{aligned}
d\widetilde{X}_t^{\boldsymbol{\pi}} = {} & \sigma\hat{\rho}_t \left(\int_{\mathbb{R}} u \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du\right) dt + \sigma \left(\int_{\mathbb{R}} u \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du\right) d\hat{B}_t \\
& + \sigma \sqrt{\int_{\mathbb{R}} u^2 \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du - \left(\int_{\mathbb{R}} u \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du\right)^2} d\overline{B}_t,
\end{aligned} \tag{12}$$

where an additional Brownian motion $\overline{B}$ independent of $\hat{B}$, adapted to $\{\overline{\mathcal{F}}_t : t \in [0,T]\}$, is introduced due to the randomness caused by exploration. The sample state process $\{X_t^{\boldsymbol{\pi}} : t \in [0,T]\}$ is identical in law to the exploratory state process $\{\widetilde{X}_t^{\boldsymbol{\pi}} : t \in [0,T]\}$ as pointed out in Jia and Zhou (2022b). Thus, the RSEMV problem (11) is equivalent to

$$\min_{\boldsymbol{\pi}\in\Pi} \mathbb{E}\left[\left(\widetilde{X}_T^{\boldsymbol{\pi}} - \omega\right)^2 - \Lambda \int_0^T \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t)\right) dt\right] - (\omega - z)^2, \tag{13}$$

where the expectation is taken with respect to the randomness of $\hat{B}$ and $\overline{B}$ only. Compared with the EMV problem studied in Wang and Zhou (2020) for the market without regime switching, our problem is more challenging due to the presence of the additional state p_t.

We now specify Π, the set of admissible feedback control distributions. A policy $\boldsymbol{\pi} = \boldsymbol{\pi}(\cdot \mid \cdot, \cdot, \cdot)$ is called admissible if

(i) $\boldsymbol{\pi}(\cdot \mid t, x, p) \in \mathcal{P}(\mathbb{R})$ for any $(t, x, p) \in [0,T] \times \mathbb{R} \times (0,1)$, and $\boldsymbol{\pi}(\cdot \mid t, x, p) : (t, x, p, u) \in [0,T] \times \mathbb{R} \times (0,1) \times \mathbb{R} \mapsto \mathbb{R}$ is measurable;

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(ii) The SDE (12) admits a unique strong solution for any initial state $(t, x, p) \in [0, T] \times \mathbb{R} \times (0, 1)$;

(iii) $\mathbb{E}_{t,x,p}\left[\int_t^T \int_{\mathbb{R}} u^2 \boldsymbol{\pi}\left(u \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s\right) du ds\right]$ is finite for any $(t, x, p) \in [0, T] \times \mathbb{R} \times (0, 1)$;

(iv) $\mathbb{E}_{t,x,p}\left[\left(\widetilde{X}_T^{\boldsymbol{\pi}} - \omega\right)^2 + \left|\Lambda \int_t^T \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s)\right) ds\right|\right]$ is finite for any $(t, x, p) \in [0, T] \times \mathbb{R} \times (0, 1)$.

Here, $\mathbb{E}_{t,x,p}[\cdot]$ denotes taking expectation under $\mathbb{P}$ given $\widetilde{X}_t^{\boldsymbol{\pi}} = x$ and $p_t = p$. Condition (iv) ensures finiteness of the objective.

To solve the optimal control problem (13), we define the value function associated with policy $\boldsymbol{\pi}$ as

$$V^{\boldsymbol{\pi}}(t, x, p) := \mathbb{E}_{t,x,p}\left[\left(\widetilde{X}_T^{\boldsymbol{\pi}} - \omega\right)^2 - \Lambda \int_t^T \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s)\right) ds\right] - (\omega - z)^2.$$

The optimal value function is defined as

$$V^*(t, x, p) := \min_{\boldsymbol{\pi} \in \Pi} V^{\boldsymbol{\pi}}(t, x, p).$$

3 Solution to the RSEMV Problem

We solve the RSEMV problem (13) by dynamic programming. Following standard arguments, we deduce that the optimal value function satisfies the Hamilton–Jacobi–Bellman (HJB) equation

$$v_t + \min_{\boldsymbol{\pi} \in \Pi} \int_{\mathbb{R}} \left(\frac{1}{2} v_{xx} \sigma^2 u^2 + [(\rho_2 + (\rho_1 - \rho_2)p) v_x + (\rho_1 - \rho_2) p (1 - p) v_{xp}] u\sigma + \Lambda \ln \boldsymbol{\pi}(u)\right) \boldsymbol{\pi}(u) du$$

$$+ \frac{1}{2} (\rho_1 - \rho_2)^2 p^2 (1 - p)^2 v_{pp} + [\lambda_2 - (\lambda_1 + \lambda_2) p] v_p = 0,$$

with the terminal condition $v(T, x, p) = (x - \omega)^2 - (\omega - z)^2$. Here, v is the generic unknown solution to the HJB equation, and v_t indicates the partial derivative of v with respect to t and similary for other partial derivatives.

To solve the minimization problem in the HJB equation, consider the function

$$\left(\frac{1}{2} v_{xx} \sigma^2 u^2 + [(\rho_2 + (\rho_1 - \rho_2)p) v_x + (\rho_1 - \rho_2) p (1 - p) v_{xp}] u\sigma + \Lambda \ln y\right) y,$$

which has a unique minimizer

$$y^* = \exp\left(-\frac{1}{\Lambda}\left(\frac{1}{2} v_{xx} \sigma^2 u^2 + [(\rho_2 + (\rho_1 - \rho_2)p) v_x + (\rho_1 - \rho_2) p (1 - p) v_{xp}] u\sigma\right)\right)$$

Thus, the density of the probability distribution that minimizes the integral is given by

$$\begin{aligned}\boldsymbol{\pi}^*(u \mid t, x, p) &= \frac{\exp\left(-\frac{1}{\Lambda}\left(\frac{1}{2} v_{xx} \sigma^2 u^2 + [(\rho_2 + (\rho_1 - \rho_2)p) v_x + (\rho_1 - \rho_2) p (1 - p) v_{xp}] u\sigma\right)\right)}{\int_{\mathbb{R}} \exp\left(-\frac{1}{\Lambda}\left(\frac{1}{2} v_{xx} \sigma^2 u^2 + [(\rho_2 + (\rho_1 - \rho_2)p) v_x + (\rho_1 - \rho_2) p (1 - p) v_{xp}] u\sigma\right)\right) du} \\ &= \mathcal{N}\left(u \mid -\frac{(\rho_2 + (\rho_1 - \rho_2)p) v_x + (\rho_1 - \rho_2)\, p\, (1 - p)\, v_{xp}}{\sigma v_{xx}}, \frac{\Lambda}{\sigma^2 v_{xx}}\right), \end{aligned} \tag{14}$$

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provided $v_{xx} > 0$, where $\mathcal{N}(\cdot \mid m, \text{var})$ is the density of normal distribution with mean m and variance var. Using the expression of $\boldsymbol{\pi}^*(u \mid t, x, p)$, the HJB equation becomes

$$v_t - \frac{\left[(\rho_2 + (\rho_1 - \rho_2)p)v_x + (\rho_1 - \rho_2)\, p\,(1-p)\, v_{xp}\right]^2}{2v_{xx}} - \frac{\Lambda}{2}\ln\frac{2\pi\Lambda}{\sigma^2 v_{xx}} + \frac{1}{2}(\rho_1 - \rho_2)^2 p^2 (1-p)^2 v_{pp} + \left[\lambda_2 - (\lambda_1 + \lambda_2)\, p\right] v_p = 0. \tag{15}$$

To solve this nonlinear PDE, we adopt the ansatz

$$v(t, x, p) = (x - \omega)^2 \frac{1}{f(t, p)} + g(t, p) - (\omega - z)^2, \tag{16}$$

which separates x from t, p. Plugging it back to (15), we obtain that $f(t, p)$ solves

$$\begin{aligned} &f_t + \frac{1}{2}(\rho_1 - \rho_2)^2 p^2 (1-p)^2 f_{pp} - \left[2(\rho_2 + (\rho_1 - \rho_2)p)(\rho_1 - \rho_2)\, p\,(1-p) + (\lambda_1 + \lambda_2)\, p - \lambda_2\right] f_p \\ &\quad + (\rho_2 + (\rho_1 - \rho_2)p)^2 f = 0,\ (t, p) \in [0, T) \times (0, 1), \\ &f(T, p) = 1, p \in (0, 1). \end{aligned} \tag{17}$$

In addition, $g(t, p)$ solves

$$\begin{aligned} &g_t + \frac{1}{2}(\rho_1 - \rho_2)^2 p^2 (1-p)^2 g_{pp} + \left[\lambda_2 - (\lambda_1 + \lambda_2)\, p\right] g_p - \frac{\Lambda}{2}\ln\frac{\pi\Lambda f}{\sigma^2} = 0, (t, p) \in [0, T) \times (0, 1), \\ &g(T, p) = 0,\ p \in (0, 1). \end{aligned} \tag{18}$$

Note that $f(t, p)$ appears in the source term of the PDE for $g(t, p)$ if $\Lambda > 0$.

It is worth emphasizing that the two PDEs (17) and (18) do not require boundary conditions for their solutions to be unique. This is because 0 and 1 are both entrance boundaries for the filtered regime probability, which cannot be hit when the initial value $p_0 \in (0, 1)$; see Ekström and Tysk (2011) for detailed discussions in the context of term structure models that have the same boundary behavior.

Both PDEs are degenerate in the sense that the coefficient function before f_{pp} or g_{pp} becomes zero at $p = 0, 1$. Thus, we cannot apply standard results for linear parabolic PDEs to infer properties of $f(t, p)$ and $g(t, p)$, which requires the PDEs to be uniformly parabolic (i.e., the coefficient functions before f_{pp} and g_{pp} should be bounded away from zero for any $p \in (0, 1)$). We use the result in Friedman (1976) for degenerate linear parabolic PDEs to establish the existence of the solutions to (17) and (18), and the maximum principle to show uniqueness and obtain some nice properties. Our results are summarized in the following theorem.

Theorem 1. *The following properties hold for solutions to PDEs (17) and (18).*
(1) PDE (17) admits a unique solution $f \in C^{1,2}([0, T) \times [\varepsilon, 1-\varepsilon]) \cap C([0, T] \times [\varepsilon, 1-\varepsilon])$ for any $\varepsilon \in (0, 1/2)$. Furthermore, we have $1 \le f(t, p) \le e^{\overline{\rho}^2 (T-t)}$ and $|f_p(t, p)| \le K$ in $[0, T] \times (0, 1)$, where $\overline{\rho} = \max\{|\rho_1|, |\rho_2|\}$ and K is a positive constant. Consequently, $v_{xx}(t, x, p) = 2/f(t, p) > 0$, where $v(t, x, p)$ is given by (16). Moreover, for any $p \in (0, 1)$, $f(t, p)$ is a decreasing function of t.
(2) PDE (18) admits a unique solution $g \in C^{1,2}([0, T) \times [\varepsilon, 1-\varepsilon]) \cap C([0, T] \times [\varepsilon, 1-\varepsilon])$ for any $\varepsilon \in (0, 1/2)$. Furthermore, we have $c(T-t) \le g(t, p) \le C(T-t)$ in $[0, T] \times (0, 1)$, where c, C are two constants.

We do not have analytical solutions to (17) and (18). It is difficult to obtain analytical solutions to these PDEs as the coefficient functions of f_{pp} and g_{pp} are fourth-order polynomials of p. Nevertheless, once the environment parameters are given, they can be solved by standard numerical methods such as finite difference. As these two PDEs are not coupled, we can first solve for $f(t, p)$ numerically and then solve for $g(t, p)$ numerically based on the numerical solution

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of $f(t,p)$. However, it should be noted that, in our RL algorithm, we do not solve these PDEs numerically to obtain approximations of $f(t,p)$ and $g(t,p)$.

Now, we provide a verification theorem to show that the optimal value function is indeed given by the ansatz (16) and the optimal feedback distribution has density given by (14).

Theorem 2. *The optimal value function of the unconstrained MV problem* (13) *with parameter* ω *is given by*

$$V^*(t,x,p) = (x-\omega)^2 \frac{1}{f(t,p)} + g(t,p) - (\omega - z)^2, \tag{19}$$

for $(t,x,p) \in [0,T] \times \mathbb{R} \times (0,1)$*, and the optimal feedback distribution is Gaussian with density function*

$$\boldsymbol{\pi}^*(u \mid t,x,p) = \mathcal{N}\left(u \mid m^*(t,x,p), \frac{f(t,p)\Lambda}{2\sigma^2}\right), \tag{20}$$

where

$$m^*(t,x,p) = -\frac{x-\omega}{\sigma}\left(\rho_2 + (\rho_1-\rho_2)p - (\rho_1-\rho_2)\,p(1-p)\frac{f_p(t,p)}{f(t,p)}\right), \tag{21}$$

and the functions $f(t,p)$ *and* $g(t,p)$ *satisfy PDE* (17) *and* (18)*, respectively.*

Theorem 2 shows that the optimal exploration distribution is Gaussian in our problem, which is similar to the case without regime switching studied in Wang and Zhou (2020). However, unlike their problem, now the filtered regime probability affects the mean of the exploration distribution in addition to the discounted wealth.

At time t, given the states x and p, the mean of our control distribution can be decomposed into two parts

$$-\frac{x-\omega}{\sigma}\left(\rho_2 + (\rho_1-\rho_2)p\right) + \frac{x-\omega}{\sigma}\left(\rho_1-\rho_2\right)p(1-p)\frac{f_p(t,p)}{f(t,p)}$$

In the first component, the term $\rho_2 + (\rho_1-\rho_2)p$ is the current expected value of the Sharpe ratio of the risky asset based on the filtered regime probability. Thus, it can be viewed as myopic as it only considers the current state. The second component further considers the drift risk caused by possible switches in the regimes and reflects hedging demand. A similar decomposition of the optimal control is obtained in Rieder and Bäuerle (2005) for utility maximization in the regime-switching market.

The variance of $\boldsymbol{\pi}^*$ represents the optimal extent of exploration, which is proportional to Λ and $f(t,p)$ and inversely proportional to σ^2. All other things being equal, a larger Λ or a smaller volatility leads to a larger variance for exploration. Furthermore, with the same Λ and σ^2, the extent of exploration decays as time approaches the end date of the investment horizon, as seen from the decreasing property of $f(t,p)$ as a function of t with fixed p. These observations are consistent with the findings in Wang and Zhou (2020) for the market without regime switching.

Remark 2. If there is only one regime in the market, $m^*(t,x,p)$ becomes $-(x-w)\rho/\sigma$, where ρ is the Sharpe ratio of the risky asset. Thus, the mean of the optimal control distribution reduces to the one obtained in Wang and Zhou (2020) for the market without regime switching.

As a corollary to Theorem 2, we establish the equivalence between the classical MV problem in the two-regime market (3) without exploration and the RSEMV problem as the exploration weight Λ goes to 0.

Corollary 1. *For the unconstrained MV problem* (2) *without exploration, its optimal value function is given by*

$$V^*_{\mathrm{cl}}(t,x,p) = (x-\omega)^2 \frac{1}{f(t,p)} - (\omega - z)^2, \tag{22}$$

and the optimal feedback control is given by $m^*(t,x,p)$*.*

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As $\Lambda \to 0$, $V^*(t,x,p)$ given by (19) converges pointwisely to $V^*_{\text{cl}}(t,x,p)$, and the optimal feedback policy $\boldsymbol{\pi}^*(\cdot \mid t,x,p)$ given by (20) converges weakly to $\delta_{m^*(t,x,p)}(\cdot)$, the Dirac measure concentrated at $\delta_{m^*(t,x,p)}(\cdot)$.

Remark 3. Note that $m^*(t,x,p)$ depends on ω. The optimal feedback control of the constrained MV problem (1) can be obtained by setting $\omega = \omega^*$ (the optimal Lagrange multiplier) in the formula of $m^*(t,x,p)$.

Using Theorem 2 in Jia and Zhou (2022c), we obtain a policy improvement result. Specifically, if $\boldsymbol{\pi}$ is an admissible feedback control policy and $V^{\boldsymbol{\pi}}$ is the value function of $\boldsymbol{\pi}$, let

$$\boldsymbol{\pi}'(u \mid t,x,p) := \mathcal{N}\left(u \mid -\frac{(\rho_2 + (\rho_1-\rho_2)p)V^{\boldsymbol{\pi}}_x + (\rho_1-\rho_2)\,p\,(1-p)\,V^{\boldsymbol{\pi}}_{xp}}{\sigma V^{\boldsymbol{\pi}}_{xx}}, \frac{\Lambda}{\sigma^2 V^{\boldsymbol{\pi}}_{xx}}\right). \tag{23}$$

Suppose $\boldsymbol{\pi}'$ is admissible. Then, $V^{\boldsymbol{\pi}'}(t,x,p) \le V^{\boldsymbol{\pi}}(t,x,p)$. This suggests that one can always find a policy in the Gaussian family that improves the value function of a general policy. Thus, we only need to consider Gaussian policies in our learning. However, the policy update given by (23) cannot be exactly implemented because the partial derivatives of $V^{\boldsymbol{\pi}}$ involve unknown environment parameters.

4 RL Algorithm Design

The representation of the optimal policy obtained in Theorem 2 is not implementable because the environment parameters are unknown. In this section, based on the semi-analytical representations we derived for the optimal value function and optimal policy as well as the general theory of RL in continuous time developed in Jia and Zhou (2022a,b), we develop an actor-critic RL algorithm, named partially observable exploratory mean-variance (POEMV) algorithm, to learn the optimal policy.

4.1 Parametrization of Value Function and Policy

Given the representation of the optimal value function given by (19), it is natural for us to introduce the following parametrization for the value function:

$$V^\theta(t,x,p) = (x-\omega)^2 \frac{1}{f^\theta(t,p)} + g^\theta(t,p) - (\omega - z)^2, \tag{24}$$

where

$$f^\theta(t,p) = \exp\left(\sum_{0\le i\le m, 1\le j\le m} \theta_{i,j} p^i (T-t)^j\right), \tag{25}$$

$$g^\theta(t,p) = \sum_{0\le i\le m, 1\le j\le m} \tilde{\theta}_{i,j} p^i (T-t)^j, \tag{26}$$

$$\theta = (\theta_{0,1}, \theta_{1,1}, \ldots, \theta_{m,m}, \tilde{\theta}_{0,1}, \tilde{\theta}_{1,1}, \ldots, \tilde{\theta}_{m,m}) \in \mathbb{R}^{2m(m+1)}.$$

The two functions $f(t,p)$ and $g(t,p)$ are solutions to the PDEs (17) and (18). However, we do not approximate them by numerically solving these PDEs because the true environment parameters are unknown. Instead, we approximate $\ln f(t,p)$ and $g(t,p)$ by a polynomial of order m and learn the coefficients in the RL algorithm.

Substituting the parametrized value function back to (14) yields

$$\mathcal{N}\left(u \mid -(x-\omega)\left[\frac{\rho_1-\rho_2}{\sigma}\left(p - \frac{f^\theta_p(t,p)}{f^\theta(t,p)} p(1-p)\right) + \frac{\rho_2}{\sigma}\right], \frac{\Lambda}{2\sigma^2} f^\theta(t,p)\right).$$

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Considering the above form, we parametrize our policy as

$$\boldsymbol{\pi}^{\phi}\left(u \mid t, x, p\right) = \mathcal{N}\left(u \mid -(x-\omega)\left[\phi_1\left(p - \frac{f_p^{\theta}(t,p)}{f^{\theta}(t,p)}p(1-p)\right) + \phi_2\right], \Lambda e^{\phi_3} f^{\theta}(t,p)\right), \tag{27}$$

and denote its mean by μ^{ϕ} and variance by σ^{ϕ}. We also introduce the vector $\phi = (\phi_1, \phi_2, \phi_3) \in \mathbb{R}^3$, which contains all the policy parameters. We use e^{ϕ_3} to parametrize the variance of the policy to avoid getting negative values for the variance in the training process. We also separate Λ out in the variance instead of merging it with e^{ϕ_3} because it is a hyperparameter set by the agent and not a parameter to be learned. For $\boldsymbol{\pi}^{\phi}$ in (27), its differential entropy is given by

$$\mathcal{H}\left(\boldsymbol{\pi}^{\phi}\left(\cdot \mid t, x, p\right)\right) = \frac{1}{2}\ln\left(2\pi e \sigma^{\phi}\right),$$

which does not depend on x.

One can also parametrize the value function and policy in other ways, e.g., using neural networks, as done in popular model-free RL algorithms, such as DDPG (Lillicrap et al. (2015)) and SAC (Haarnoja et al. (2018)) for problems with continuous states and actions. However, neural networks typically involve a great number of parameters, which can make learning slow. Furthermore, such algorithms based on neural networks can suffer from convergence issues like instability in training and lead to unsatisfactory out-of-sample results; see Wang and Zhou (2020), Huang et al. (2022), and Wang et al. (2023). In contrast, our parametrizations based on the semi-analytical representations of the optimal value function and policy only require a few parameters, making learning potentially easier.

4.2 Interaction and Filtering

In RL, the agent interacts with the environment repeatedly to update her guess of the policy and its value function. Although our problem is in continuous time, we can only interact with the environment in discrete time with a given policy $\boldsymbol{\pi}$. We consider a time grid $\{t_0, \cdots, t_K\}$ with $t_k = k\Delta t$ and $\Delta t = T/K$. At each t_k, we observe S_{t_k}, the price of the risky asset, and $X_{t_k}^{\boldsymbol{\pi}}$, the discounted value of our wealth under policy $\boldsymbol{\pi}$. As the regime is unobservable, we perform filtering to obtain p_t, the belief probability of the regime at time t, based on S_t and past prices. Specifically, we assume that we have already estimated the environment parameters from some historical data and denote the estimates by $\tilde{\sigma}, \tilde{\mu}_i, \tilde{\lambda}_i$ for $i = 1, 2$. We also calculate $\tilde{\rho}_i$ using $\tilde{\sigma}$ and $\tilde{\mu}_i$. At time t_k, we first discretize time in the innovation process (7) and obtain its increment

$$\Delta\hat{B}_{t_k} = \hat{B}_{t_k} - \hat{B}_{t_{k-1}} = \frac{1}{\tilde{\sigma}}\left[\log\left(\frac{S_{t_k}}{S_{t_{k-1}}}\right) - \left(\hat{\mu}_{t_{k-1}} - \tilde{\sigma}^2/2\right)\Delta t\right], \tag{28}$$

where $\hat{\mu}_{t_{k-1}} = \tilde{\mu}_1 p_{t_{k-1}} + \tilde{\mu}_2(1 - p_{t_{k-1}})$. We then obtain p_{t_k} from the Wonham filter (10). Directly applying the Euler scheme for (10) can generate values outside $[0, 1]$, which are not valid probabilities. To avoid such issue, we consider the transformation $q_t^1 := \log p_t$ and $q_t^2 := \log(1 - p_t)$ and obtain the SDEs of q_t^1 and q_t^2 by Ito's formula. We then apply the Euler scheme to these SDEs and obtain

$$\begin{aligned} q_{t_k}^1 = q_{t_{k-1}}^1 &+ \left[\tilde{\lambda}_2 e^{-q_{t_{k-1}}^1} - \left(\tilde{\lambda}_1 + \tilde{\lambda}_2\right) - \frac{1}{2}\left(\tilde{\rho}_1 - \tilde{\rho}_2\right)^2\left(1 - e^{q_{t_{k-1}}^1}\right)^2\right]\Delta t \\ &+ \left(\tilde{\rho}_1 - \tilde{\rho}_2\right)\left(1 - e^{q_{t_{k-1}}^1}\right)\Delta\hat{B}_{t_k}, \end{aligned} \tag{29}$$

$$\begin{aligned} q_{t_k}^2 = q_{t_{k-1}}^2 &+ \left[\left(\tilde{\lambda}_1 + \tilde{\lambda}_2\right)\left(e^{-q_{t_{k-1}}^2} - 1\right) - \tilde{\lambda}_2 e^{-q_{t_{k-1}}^2} - \frac{1}{2}\left(\tilde{\rho}_1 - \tilde{\rho}_2\right)^2\left(1 - e^{q_{t_{k-1}}^2}\right)^2\right]\Delta t \\ &- \left(\tilde{\rho}_1 - \tilde{\rho}_2\right)\left(1 - e^{q_{t_{k-1}}^2}\right)\Delta\hat{B}_{t_k}. \end{aligned} \tag{30}$$

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We calculate p_{t_k} as

$$p_{t_k} = \frac{e^{q^1_{t_k}}}{e^{q^1_{t_k}} + e^{q^2_{t_k}}}, \tag{31}$$

which is guaranteed to be a value in $(0, 1)$. Excellent performance of this approximation scheme was demonstrated in Yin et al. (2004).

Given $(X^{\boldsymbol{\pi}}_{t_k}, p_{t_k})$, we sample action u_{t_k} from the distribution $\boldsymbol{\pi}(\cdot \mid t_k, X^{\boldsymbol{\pi}}_{t_k}, p_{t_k})$ and apply it. The environment then generates $S_{t_{k+1}}$ and $X^{\boldsymbol{\pi}}_{t_{k+1}}$ for time t_{k+1}, where

$$X^{\boldsymbol{\pi}}_{t_{k+1}} = X^{\boldsymbol{\pi}}_{t_k} + u_{t_k} \frac{S_{t_{k+1}} - S_{t_k}}{S_{t_k}}.$$

Continuing this procedure until time reaches T, we obtain a full trajectory of states under the stochastic policy $\boldsymbol{\pi}$, $\{(t_k, X^{\boldsymbol{\pi}}_{t_k}, p_{t_k}, u_{t_k}, X^{\boldsymbol{\pi}}_{t_{k+1}}, p_{t_{k+1}}), 0, 1, \ldots, K-1\}$, which is called an episode in the RL literature.

Remark 4. An important difference between the full information case and the partial information case is that there is no need to estimate the environment parameters based on the historical data in the former one, whereas these parameters must be estimated in the latter one. In general, RL attempts to directly learn the optimal policy from interactions with the environment without estimating the environment parameters. This is possible when all the states are observable. However, in the partial information case, we must filter the unobservable states for learning, which requires estimates of the environment.

4.3 Policy Evaluation (PE)

Jia and Zhou (2022a) developed a martingale theory for PE in continuous-time problems. We proposed two types of methods for PE. One is minimizing the martingale loss (ML) function, which results in an offline[1] algorithm, and the other is solving the martingale orthogonality condition, which leads to online algorithms such as different versions of the temporal difference algorithm. Here, we use the ML function to develop an offline algorithm, but we can also develop online algorithms following Jia and Zhou (2022a).

For any policy $\boldsymbol{\pi}$,

$$M_t := V^{\boldsymbol{\pi}}\left(t, X^{\boldsymbol{\pi}}_t, p_t\right) - \Lambda \int_0^t \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid s, X^{\boldsymbol{\pi}}_s, p_s)\right) ds$$

is a martingale with respect to $\{\overline{\mathcal{F}}_t : t \in [0, T]\}$. To find the value function parameter vector θ for a given policy $\boldsymbol{\pi}^\phi$, we consider

$$M^\theta_t := V^\theta\left(t, X^{\boldsymbol{\pi}^\phi}_t, p_t\right) - \Lambda \int_0^t \mathcal{H}\left(\boldsymbol{\pi}^\phi(\cdot \mid s, X^{\boldsymbol{\pi}^\phi}_s, p_s)\right) ds.$$

Then, we minimize the ML function

$$\begin{aligned} ML(\theta; \phi) &:= \frac{1}{2}\overline{\mathbb{E}} \int_0^T \left| M_T - M^\theta_t \right|^2 \, \mathrm{d}t \\ &= \frac{1}{2}\overline{\mathbb{E}} \int_0^T \left[h\left(X^{\boldsymbol{\pi}^\phi}_T\right) - V^\theta\left(t, X^{\boldsymbol{\pi}^\phi}_t, p_t\right) - \Lambda \int_t^T \mathcal{H}\left(\boldsymbol{\pi}^\phi(\cdot \mid s, X^{\boldsymbol{\pi}^\phi}_s, p_s)\right) ds \right]^2 dt, \end{aligned}$$

[1] An offline algorithm updates the parameters when an episode ends, whereas an online algorithm updates the parameters at every time step of the episode; see e.g., Sutton and Barto (2018).

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where $h(x) := (x-w)^2 - (w-z)^2$. The minimizer is given by the true parameter vector of the value function of $\boldsymbol{\pi}^\phi$. As we can only obtain episodes in discrete time, we approximate the ML function by

$$
\begin{aligned}
ML(\theta;\phi) &\approx ML_{\Delta t}(\theta;\phi) \\
&:= \frac{1}{2}\overline{\mathbb{E}}\sum_{k=0}^{K-1}\left[h\left(X_T^{\boldsymbol{\pi}^\phi}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right) - \sum_{i=k}^{K-1}\Lambda\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_i, X_{t_i}^{\boldsymbol{\pi}^\phi}, p_{t_i}\right)\right)\Delta t\right]^2 \Delta t.
\end{aligned}
$$

Then, we can approximate the gradient of the ML function as

$$
\begin{aligned}
\frac{\partial}{\partial\theta}ML(\theta;\phi) &\approx \frac{\partial}{\partial\theta}ML_{\Delta t}(\theta;\phi) \\
&= -\overline{\mathbb{E}}\sum_{k=0}^{K-1}\left[\left(h\left(X_T^{\boldsymbol{\pi}^\phi}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right) - \sum_{i=k}^{K-1}\Lambda\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_i, X_{t_i}^{\boldsymbol{\pi}^\phi}, p_{t_i}\right)\right)\Delta t\right)\right. \\
&\quad \left.\times\frac{\partial V^\theta}{\partial\theta}\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right]\Delta t,
\end{aligned}
$$

where

$$
\frac{\partial V^\theta}{\partial\theta_{i,j}}(t,x,p) = -(x-\omega)^2\frac{p^i(T-t)^j}{f^\theta(t,x,p)}, \tag{32}
$$

$$
\frac{\partial V^\theta}{\partial\tilde{\theta}_{i,j}}(t,x,p) = p^i(T-t)^j, \tag{33}
$$

for $i = 0, 1..., m$, $j = 1..., m$. To find the minimizer of the ML function, we use gradient descent and update θ as

$$
\theta \leftarrow \theta - \eta_\theta\frac{\partial}{\partial\theta}ML_{\Delta t}(\theta;\phi), \tag{34}
$$

where $\eta_\theta > 0$ is the learning rate for θ. To save computations, we only update θ once in our implementation for the given policy instead of performing it many times to get close to the true θ for this policy. This practice is often adopted for actor-critic algorithms; see Sutton and Barto (2018).

4.4 Policy Gradient (PG)

We update the policy using gradient descent, which requires knowing the gradient of the value function $G(\phi) := \frac{\partial}{\partial\phi}V^{\boldsymbol{\pi}^\phi}(0,x,p)$ for $X_0^{\boldsymbol{\pi}^\phi} = x$ and $p_0 = p$.

Jia and Zhou (2022b) showed that the PG problem can be converted into a PE problem. Applying Theorem 5 in their paper, we have

$$
\begin{aligned}
G(\phi) =& \overline{\mathbb{E}}\left\{\int_0^T \frac{\partial}{\partial\phi}\ln\boldsymbol{\pi}^\phi\left(u_s^{\boldsymbol{\pi}^\phi} \mid s, X_s^{\boldsymbol{\pi}^\phi}, p_s\right)\left[dV^{\boldsymbol{\pi}^\phi}\left(s, X_s^{\boldsymbol{\pi}^\phi}, p_s\right) - \Lambda\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid s, X_s^{\boldsymbol{\pi}^\phi}, p_s\right)\right)ds\right]\right. \\
&\left. -\Lambda\frac{\partial}{\partial\phi}\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid s, X_s^{\boldsymbol{\pi}^\phi}, p_s\right)\right)ds\right\},
\end{aligned}
$$

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where $u_s^{\boldsymbol{\pi}^\phi}$ is the sampled action from $\boldsymbol{\pi}^\phi\left(\cdot \mid s, X_s^{\boldsymbol{\pi}^\phi}, p_s\right)$ at time s. Applying time discretization and replacing $V^{\boldsymbol{\pi}^\phi}(t,x,p)$ by its parametrized function $V^\theta(t,x,p)$, we obtain

$$
\begin{aligned}
G(\phi;\theta) &\approx G_{\Delta t}(\phi;\theta) \\
&:= \overline{\mathbb{E}}\left\{\sum_{k=0}^{K-1} \frac{\partial}{\partial \phi} \ln \boldsymbol{\pi}^\phi\left(u_{t_k}^{\boldsymbol{\pi}^\phi} | t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\left[V^\theta\left(t_{k+1}, X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}, p_{t_{k+1}}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right.\right. \\
&\quad \left.\left. -\Lambda \mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right)\Delta t\right] - \Lambda \frac{\partial}{\partial \phi}\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right)\Delta t\right\},
\end{aligned}
$$

where

$$
\begin{aligned}
&\frac{\partial}{\partial \phi_n} \ln \boldsymbol{\pi}^\phi(u \mid t,x,p) = -\frac{(u-\mu^\phi)(x-\omega)}{\sigma^\phi}\left(\left[p - \frac{f_p^\theta(t,p)}{f^\theta(t,p)} p(1-p)\right]\mathbb{1}_{\{n=1\}} + \mathbb{1}_{\{n=2\}}\right), \\
&\frac{\partial}{\partial \phi_3} \ln \boldsymbol{\pi}^\phi(u \mid t,x,p) = \frac{\left(u-\mu^\phi\right)^2 - \sigma^\phi}{2\sigma^\phi}, \\
&\frac{\partial}{\partial \phi_1}\mathcal{H}\left(\boldsymbol{\pi}^\phi(\cdot \mid t,x,p)\right) = \frac{\partial}{\partial \phi_2}\mathcal{H}\left(\boldsymbol{\pi}^\phi(\cdot \mid t,x,p)\right) = 0, \\
&\frac{\partial}{\partial \phi_3}\mathcal{H}\left(\boldsymbol{\pi}^\phi(\cdot \mid t,x,p)\right) = \frac{1}{2},
\end{aligned}
$$

where μ^ϕ and σ^ϕ are given in (27). We update ϕ by gradient descent as

$$
\phi \leftarrow \phi - \eta_\phi G_{\Delta t}(\phi;\theta), \tag{35}
$$

where $\eta_\phi > 0$ is the learning rate for ϕ.

4.5 Actor-Critic Algorithm

Based on the previous results, we now develop an actor-critic algorithm for the MV problem with unobservable regimes.

Step 1: Estimation of the environment parameters. Given historical data of the market that predates the episodes that will be generated from interactions, we estimate $\tilde{\sigma}, \tilde{\mu}_i, \tilde{\lambda}_i$ for $i = 1, 2$, which can be done by in various ways; see e.g., Krishnamurthy (2016), Elliott et al. (2008), and Anandkumar et al. (2012) for algorithms for estimating hidden Markov models.

Step 2: Interacting with the environment and filtering the belief probability. Given policy $\boldsymbol{\pi}^\phi$, we collect an episode $\{(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}, u_{t_k}^{\boldsymbol{\pi}^\phi}, X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}, p_{t_{k+1}}), 0, 1, \ldots, K-1\}$ by using this policy to interact with the environment, where each p_{t_k} is calculated by the discretized Wonham filter using the estimated environment parameters from step 1; see (28), (29), (30), and (31).

Step 3: PE update. For the given policy $\boldsymbol{\pi}^\phi$ and the episode collected in step 2, we implement (34) by stochastic gradient descent (SGD) and obtain

$$
\begin{aligned}
\theta \leftarrow \theta + \eta_\theta \Bigg\{\sum_{k=0}^{K-1}\Bigg[\Bigg(h\left(X_T^{\boldsymbol{\pi}^\phi}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right) - \sum_{i=k}^{K-1}\Lambda\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_i, X_{t_i}^{\boldsymbol{\pi}^\phi}, p_{t_i}\right)\right)\Delta t\Bigg) \\
\times \frac{\partial V^\theta}{\partial \theta}(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k})\Bigg]\Delta t\Bigg\}.
\end{aligned}
$$

Step 4: PG update. Using the episode collected in step 2, we implement (35) by stochastic gradient descent (SGD) and obtain

$$
\begin{aligned}
\phi \leftarrow \phi - \eta_\phi \Bigg\{\sum_{k=0}^{K-1} \frac{\partial}{\partial \phi} \ln \boldsymbol{\pi}^\phi(u_{t_k}^{\boldsymbol{\pi}^\phi} | t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k})\left[V^\theta\left(t_{k+1}, X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}, p_{t_{k+1}}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right. \\
\left. -\Lambda\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right)\Delta t\right] - \Lambda\frac{\partial \mathcal{H}}{\partial \phi}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right)\Delta t\Bigg\}.
\end{aligned}
$$

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Here, we use θ obtained from the PE update.

Remark 5. In steps 3 and 4, we use the SGD algorithm for simplicity, which approximates the expectation involved by data from only one episode. This can make the training process noisy. To reduce noise, we can use SGD with mini-batches, i.e., we approximate the expectation by taking sample average of multiple episodes.

Repeating steps 2 to 4, we collect many episodes, and in each episode, we perform PE and PG updates alternately to improve our learning of the optimal policy and the optimal value function. If the Lagrange parameter ω is fixed in the training process, we expect to uncover the optimal solution to the unconstrained MV problem (2) with fixed ω. To solve the original MV problem (1), we also need to update ω. To this end, we follow Wang and Zhou (2020) to apply a self-correcting scheme

$$\omega \leftarrow \omega - \eta_\omega (X_T - z), \tag{36}$$

which is a stochastic algorithm to solve the equation $\overline{\mathbb{E}}[X_T] = z$ and the learning rate $\eta_\omega > 0$. This scheme can be understood as follows. Suppose the empirical terminal wealth is lower than the expected level z. In this case, the self-correcting rule would increase ω so that the mean of our strategy (27) becomes greater, guiding the terminal wealth to increase to the expected level. We implement the update for ω after a fixed number (denoted by s) of training iterations involving steps 2 to 4 are performed. To reduce variance in the updating process of ω, we also replace X_T by the average of the terminal wealth over s iterations.

5 Simulation Study

We conduct a simulation study to understand the convergence behavior of the POEMV algorithm. In the study, we specify a set of parameter values for the RS-GBM model given by (3) and (4), which is used to generate observations of the risky asset price process. With known environment parameters, we can obtain the optimal control by finite difference, which provides the ground truth for testing the convergence of our RL algorithm.

To make the specified parameter values realistic, we estimate the model from 20 years of daily data of the S&P 500 index from 09/29/1992 to 10/03/2011. The estimation is done following the heuristic method used in Dai et al. (2010, 2016). Specifically, we define a decline of at least Υ_l as a bear market and an increase of at least Υ_u as a bull market. In our implementation, we set $\Upsilon_l = 19\%$ and $\Upsilon_u = 24\%$, which are the values used in Dai et al. (2010, 2016). We first identify the bull and bear regimes from the data using these two thresholds. We then use the data in each regime to estimate μ_i, λ_i for $i = 1, 2$, where μ_i is estimated by the annualized sample average of daily returns, and λ_i is estimated as the reciprocal of the average time in years spent in regime i. For σ, it is estimated by the annualized sample standard deviation of daily returns. The estimation results are shown in Table 1, and hereafter, we regard these values as the true values of the environment parameters.

μ_1	μ_2	σ	λ_1	λ_2
0.25	−0.73	0.22	0.36	2.89

Table 1: True environment parameters for the simulation study

Below, we solve the MV problem in a 10-year investment horizon with initial wealth $x_0 = 1$ and terminal target $z = 8$. Similar target levels were used in Wang and Zhou (2020) in their studies. We set the risk-free rate $r = 0$.

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Algorithm 1 POEMV Algorithm

Require: learning rates $\eta_\theta, \eta_\phi, \eta_\omega$, initial wealth x_0, target wealth level z, investment horizon T, time step $\Delta t = T/K$, exploration weight Λ, number of iterations M, number of iterations s to update the Lagrange multiplier ω.

Estimation of environment paramters: Obtain historical data that predates the episodes that will be generated and estimate $\tilde{\sigma}, \tilde{\mu}_i, \tilde{\lambda}_i$ for $i = 1, 2$ and calculate $\tilde{\rho}_i$ using $\tilde{\sigma}$ and $\tilde{\mu}_i$ by (5).

Actor-critic learning:

Initialize θ, ϕ.

for $m = 1$ to M **do**

  for $k = 0$ to $K - 1$ **do**

    Observe S_{t_k} and $X_{t_k}^{\boldsymbol{\pi}^\phi}$ from the environment. Filter p_{t_k} by (28), (29), (30), and (31). Generate action $u_{t_k}^{\boldsymbol{\pi}^\phi}$ from the distribution $\boldsymbol{\pi}^\phi(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k})$. Observe $S_{t_{k+1}}$ and $X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}$ from the environment and filter for $p_{t_{k+1}}$.

  end for

  Obtain one episode $\{(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}, u_{t_k}^{\boldsymbol{\pi}^\phi}, X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}, p_{t_{k+1}}), 0, 1, \ldots, K-1\}$ under given $\boldsymbol{\pi}^\phi$.

  Compute

$$\Delta\theta = \sum_{k=0}^{K-1} \left[\left(h\left(X_T^{\boldsymbol{\pi}^\phi}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right) - \sum_{i=k}^{K-1} \Lambda \mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_i, X_{t_i}^{\boldsymbol{\pi}^\phi}, p_{t_i}\right)\right) \Delta t \right) \times \frac{\partial V^\theta}{\partial \theta}(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}) \right] \Delta t,$$

$$\Delta\phi = \sum_{k=0}^{K-1} \frac{\partial}{\partial \phi} \ln \boldsymbol{\pi}^\phi(u_{t_k}^{\boldsymbol{\pi}^\phi} | t_k, X_{t_k}^{\boldsymbol{\pi}_k^\phi}, p_{t_k}) \left[V^\theta\left(t_{k+1}, X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}, p_{t_{k+1}}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right) - \Lambda \mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right) \Delta t \right] - \Lambda \frac{\partial \mathcal{H}}{\partial \phi}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right) \Delta t.$$

  PE Update

  Update $\theta \leftarrow \theta + \eta_\theta \Delta\theta$.

  PG Update

  Update $\phi \leftarrow \phi - \eta_\phi \Delta\phi$.

  Update the Lagrange multiplier ω

  if $(m \bmod s) == 0$ **then**

    Update $\omega \leftarrow \omega - \eta_\omega(\frac{1}{s}\sum_{j=m-s+1}^{m} X_T^j - z)$.

  end if

end for

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5.1 Finite Difference and Polynomial Approximation

Given the true environment parameters, we solve PDEs (17) and (18) by the finite difference scheme presented in Appendix B. After obtaining the numerical approximation of $f(t,p)$ on a grid, we apply the central difference to approximate $f_p(t,p)$ on the grid. Thus, using (21), we can obtain the optimal feedback control $m^*(t,x,p)$ of the unconstrained MV problem (2) for any fixed ω. In particular, setting $\omega = \omega^*$, the optimal Lagrange multiplier, in (21) yields the optimal feedback control of the constrained MV problem (1).

Figure 1 shows the surface obtained from finite difference approximation for $f(t,p)$ and

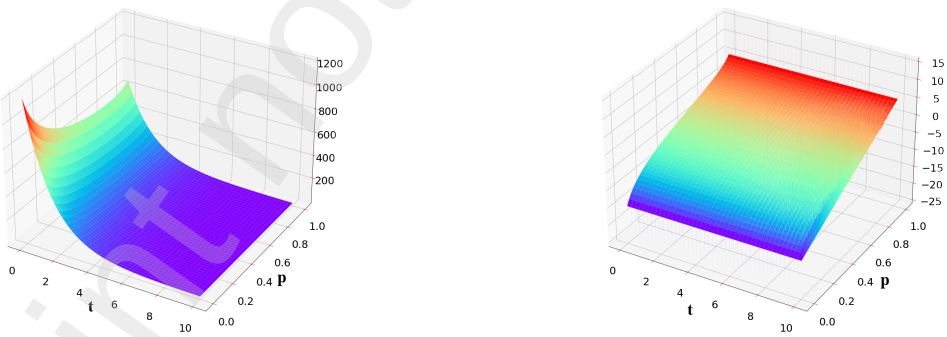
$$\gamma^*(t,p) := \frac{1}{\sigma}\left(\rho_2 + (\rho_1 - \rho_2)p - (\rho_1 - \rho_2)\, p(1-p)\frac{f_p(t,p)}{f(t,p)}\right),$$

which is part of $m^*(t,x,p)$ that is independent of $x - \omega$. It can be seen that if p is fixed, the optimal control changes quite little as t increases except near the terminal time. With t fixed, $\gamma^*(t,p)$ increases in p, which implies that the investor should invest more in the risky asset when the bull market becomes more likely if the current wealth x is below ω.

In our RL algorithm, we approximate $f(t,p)$ and $g(t,p)$ by (25) and (26), respectively. We can use the finite difference approximations of $f(t,p)$ and $g(t,p)$ as benchmarks to select the order of the polynomial. As $g(t,p)$ varies little in p with t fixed (see Figure 2(b)), we simplify our approximation of $g(t,p)$ as

$$g(t,p) \approx \sum_{1\le j\le m} \tilde{\theta}_j (T-t)^j. \tag{37}$$

We set $m = 2$ in our polynomial approximation. Figure 2 shows the finite difference surface versus the quadratic approximation surface fitted by ordinary least squares (OLS) for $\ln f(t,p)$ and $g(t,p)$. The approximation accuracy for $\ln f(t,p)$ is high when p is away from the boundary but deteriorates near the boundary. However, the error seems acceptable. For $g(t,p)$, the approximation given by (37) is highly accurate, as the difference between these two surfaces is hardly noticeable. Given these results, we use $m = 2$ in our RL algorithm.



(a) $f(t,p)$ (b) $\gamma^*(t,p)$

Figure 1: Surfaces of $f(t,p)$ and $\gamma^*(t,p)$ from finite difference

5.2 Comparison of Different Estimation Methods

Our RL algorithm requires estimating the environment before interacting with it for the purpose of filtering. Here, we compare the heuristic approach in Dai et al. (2010, 2016) and the standard EM algorithm for estimating an HMM with Gaussian noises (see e.g., Elliott et al. (2008)).

We generate 100 paths of the index over 20 years from the given set of parameters in Table 1. On each path, we apply these two estimation methods and obtain a set of estimates. We report the averaged estimates over these 100 sets in Table 2.

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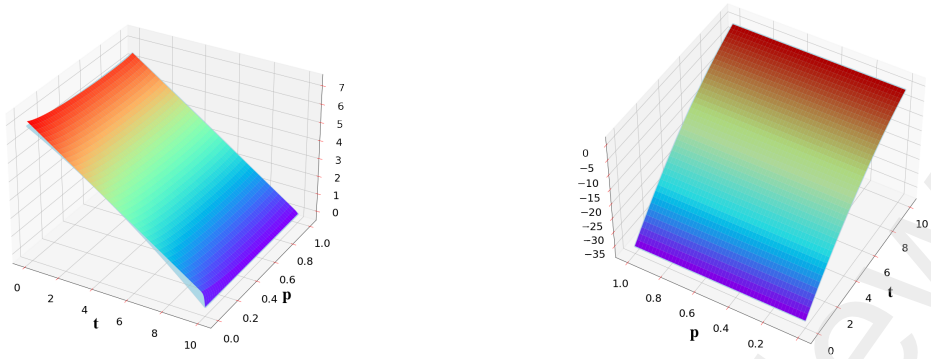

(a) $\ln f(t,p)$ (rainbow) vs. OLS fit (light blue)

(b) $g(t,p)$ (rainbow) vs. OLS fit (light blue)

Figure 2: $\ln f(t,p)$ and $g(t,p)$ from finite difference versus their OLS fits using polynomial approximation with order $m = 2$

The EM algorithm produces highly inaccurate estimates of the transition rates. The estimates of λ_1 and λ_2 are too large, implying switches as frequent as every two days, which is clearly against common sense. A similar bad performance of the EM algorithm was also reported in Sass and Haussmann (2004). In our application, the mean return is dominated by the variance on the daily level, i.e., the signal-to-noise ratio is too low. This misleads the EM algorithm to identify regime switches mainly based on the noises, which produces inaccurate estimates. In comparison, the heuristic estimation method produces good estimates for μ_2, λ_2 and σ, but overestimates μ_1 and λ_1. Due to the better performance of the heuristic approach, it will be used in further study to estimate the environment parameters.

Parameters	μ_1	μ_2	λ_1	λ_2	σ
True	0.25	-0.73	0.36	2.89	0.22
Heuristic	0.58(0.10)	-0.69(0.13)	1.21(0.42)	2.09(0.59)	0.22(0.003)
EM	0.51(0.65)	-0.21(0.28)	128.59(90.14)	132.21(88.03)	0.25(0.12)

Table 2: Comparison between the heuristic estimation method and the EM algorithm. Standard deviations of the estimates are shown in the parenthesis.

5.3 Convergence of POEMV

We analyze the convergence of the POEMV algorithm. We are interested in the convergence of ϕ_1, ϕ_2, and ω. Comparing (21) and (27), we see that if

$$\phi_1 \to \phi_1^* := \frac{\rho_1 - \rho_2}{\sigma}, \quad \phi_2 \to \phi_2^* := \frac{\rho_2}{\sigma}, \quad \omega \to \omega^*,$$

POEMV learns the optimal feedback control $m^*(t,x,p)$ for the constrained MV problem (1). The true optimal Lagrange multiplier ω^* can be computed using the self-correcting scheme (36), where we generate episodes by implementing the optimal feedback control for a given ω obtained from the finite difference scheme with the true environment parameters.

We set $\Delta t = 1/252$ and $\Lambda = 1$. We update the Lagrange multiplier ω after every $s = 10$ iterations. We initialize the parameter vectors θ and ϕ as zero vectors. For training, in each iteration, we generate a 30-year path for the index from the RS-GBM model with parameters given in Table 1, where the first 20 years of data are used for estimating the environment parameters, and the last 10 years of data are used for obtaining an episode based on the estimates of the environment parameters from the 20-year data. We train POEMV with a total of 200,000 iterations, which corresponds to using 200,000 episodes.

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We consider two settings. In one setting, we know the true environment parameters and use them in the discretized Wonham filter to obtain the belief states. In the other setting, we do not know the true environment parameters, so we use their estimated values from the heuristic approach to filter out the belief states. Comparing the results of these two settings would show how much impact the error of estimating the environment parameters would have on learning the parameters in the optimal policy.

We further assess the out-of-sample performance of POEMV in these two settings. We generate 100,000 10-year testing paths that are independent of those used for training from the RS-GBM model, and implement the learned policies on them. As a benchmark, we also implement the optimal feedback control obtained by the finite difference scheme with the true environment parameters. For filtering on each test path, we use the true environment parameters in the discretized Wonham filter for the first setting of POEMV and the benchmark, and use the estimated environment parameters (as the average of the estimated parameters over 200,000 paths) for the second setting of POEMV.

We consider three measures of out-of-sample performance: the mean and variance of the terminal wealth, and the 10-year Sharpe ratio, which is calculated by $(\text{Mean} - 1)/\sqrt{\text{Variance}}$, where 1 is the initial wealth level. We estimate the mean and variance by the sample mean and sample variance of the terminal wealth levels over 100,000 paths.

The learning process of ϕ_1 and ϕ_2 as well as the moving average of the terminal wealth level under the two settings are shown in Figure 3. The learned values for ϕ_1, ϕ_2, and ω after 200,000 iterations as well as the values of the three measures in the out-of-sample test are displayed in Table 3.

In both settings of POEMV, ω converges well to ω^*. If the true environment parameters are used for filtering, our actor-critic learning can get very close to ϕ_1^* and ϕ_2^* as seen from the second row of Table 3. If the estimated environment parameters are used for filtering, the converged results of ϕ_1 and ϕ_2 can still have a significant gap to ϕ_1^* and ϕ_2^*. Thus, the error of estimating the environment parameters, which results in the error in filtering, can impact the accuracy of learning the optimal policy. Nevertheless, despite the error in the optimal policy, the out-of-sample performance is still good. The third row of Table 3 shows that the variance of the terminal wealth is still quite small while the mean of terminal wealth meets the target.

	ϕ_1	ϕ_2	ω	Mean	Variance	Sharpe ratio
Optimal	20.25	-15.08	8.00	7.99	0.067	27.00
POEMV(true)	20.56	-15.05	8.00	7.99	0.083	24.26
POEMV(est)	15.23	-7.29	8.04	8.03	0.170	17.05

Table 3: Convergence results and out-of-sample performance. The first row shows the results of the optimal feedback control obtained by finite difference. The second and third rows show the results of POEMV using the true and estimated environment parameters for filtering, respectively.

6 Empirical Study

We conduct an empirical study to assess the performance of the POEMV algorithm on real market data. We collect 30-year daily data of the S&P500 index and 121 stocks from 09/29/1992 to 10/04/2022 from Yahoo Finance. These stocks are constituents of the index in this 30-year period and the other constituents that do not have the full 30 years of data are not considered. The S&P500 index exhibits obvious regime-switching behavior in this long time span as seen from Figure 4, where multiple bull and bear periods are observed[2].

[2]We identify these periods using the heuristic approach presented in Section 5 with $\Upsilon_l = 19\%$ and $\Upsilon_u = 24\%$.

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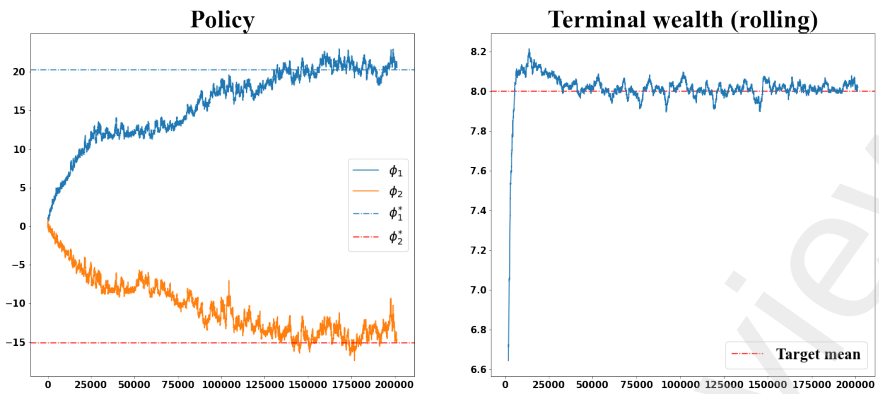
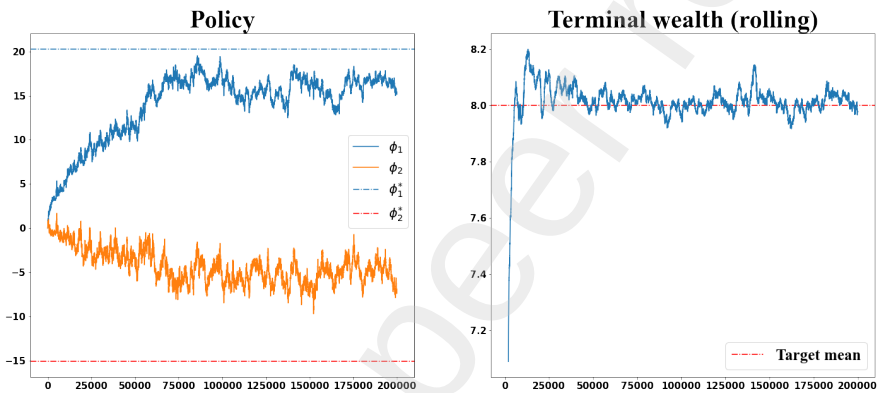


(a) POEMV with true environment parameters



(b) POEMV with estimated environment parameters

Figure 3: Convergence of POEMV for ϕ_1 and ϕ_2 and the path of terminal wealth averaged over a rolling window of 100 iterations in the training process

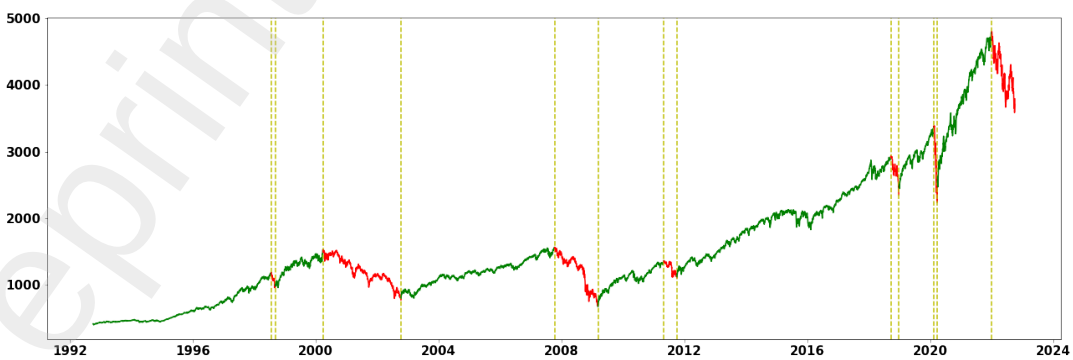


Figure 4: S&P 500 index from 09/29/1992 to 10/04/2022 with bull (green) and bear (red) periods

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Our problem is to construct a portfolio that trades the index (as an example of regime-switching assets) and the risk-free asset for the MV problem with a 10-year investment horizon with $x_0 = 1$ and $z = 8$. We set the risk-free rate $r = 0$. We split the data into the first 20 years and the last 10 years, where the former is used for training and the latter is reserved for out-of-sample testing. Unlike the simulation study where we can generate unlimited paths for training our algorithm, we only have a single trajectory observed for the index. To address the data insufficiency issue, we adopt the idea of universal training used in Wang and Zhou (2020), which treats the paths of different stocks as paths of the same stock. For our problem, we estimate the beta of each stock in our list using the first 20 years of data and select those with estimated beta between 0.9 to 1.1, which gives us 30 stocks. We treat the paths of these 30 stocks as paths of the index because their beta is close to one, implying their paths are quite close to the index path in the same period.

We estimate the environment parameters using the heuristic approach presented in Section 5 based on the first 20 years of the observed index path. Then, we apply the discretized Wonham filter with the estimated environment parameters on the 20-year path of each of these 30 stocks to obtain the belief states. Now, we have 30 paths of 20 years with filtered belief states on each day. We divide these data into overlapping 10-year blocks, which give us about 75,000 such blocks. In each iteration of our training, we randomly sample one block to apply our policy to obtain a 10-year episode for PE and PG updates.

We compare three algorithms: POEMV, the EMV algorithm of Jia and Zhou (2022b), and the classical control approach. In the EMV algorithm, the risky asset price follows a GBM model without regime-switching, so no filtering is needed. The algorithm parametrizes the value function and policy using the analytical formulas obtained in Wang and Zhou (2020) and performs actor-critic learning. The data for training the EMV algorithm is generated in the same way as the data for training POEMV. We train both algorithms with $\Delta t = 1/252$ until convergence is observed. In the classical control approach, we first estimate the environment parameters from the 20-year path of the index and then solve for the optimal control by finite difference using these estimates as if they were the true values.

In out-of-sample testing, we implement the learned policies from three algorithms on a daily and monthly basis on 122 paths, which are the paths of 121 stocks plus the index observed in the out-of-sample period. For POEMV and the classical control approach, we use the discretized Wonham filter to obtain the belief states based on the environment parameters estimated from the first 20 years of the index data. The mean and variance of the terminal wealth together with the 10-year Sharpe ratio are displayed in Table 4. The out-of-sample paths of the wealth process from POEMV and EMV are shown in Figure 5.

The mean value of the terminal wealth under all three algorithms is close to the target level of 8. However, they differ substantially in the variance of terminal wealth. The POEMV achieves the smallest among all, which is about half of the variance of EMV. The classical approach generates a large variance, which is mainly due to extremely low wealth levels observed from some stock paths. As the classical approach and our RL algorithm use the same set of estimated environment parameters for filtering, this comparison highlights the essential difference between these two approaches in obtaining the optimal policy. The RL algorithm provides a mechanism to reduce the bias in the optimal policy caused by inaccurate estimation of the environment through repeated interactions, while the classical approach only relies on the inaccurate estimates of the environment for control. As a result, the optimal policy from the classical approach can lead to significantly inferior results in some scenarios.

The comparison between POEMV and EMV underscores the importance of using the belief probability of the market regime as a signal for long-term investment. Although the policies of both algorithms experience similar ups and downs on the testing paths, it is evident from Figure 5 that the POEMV policy allows the investor to better ride the bull wave (i.e., faster growth in the bull market) and better shield the portfolio from market plunges (i.e., smaller

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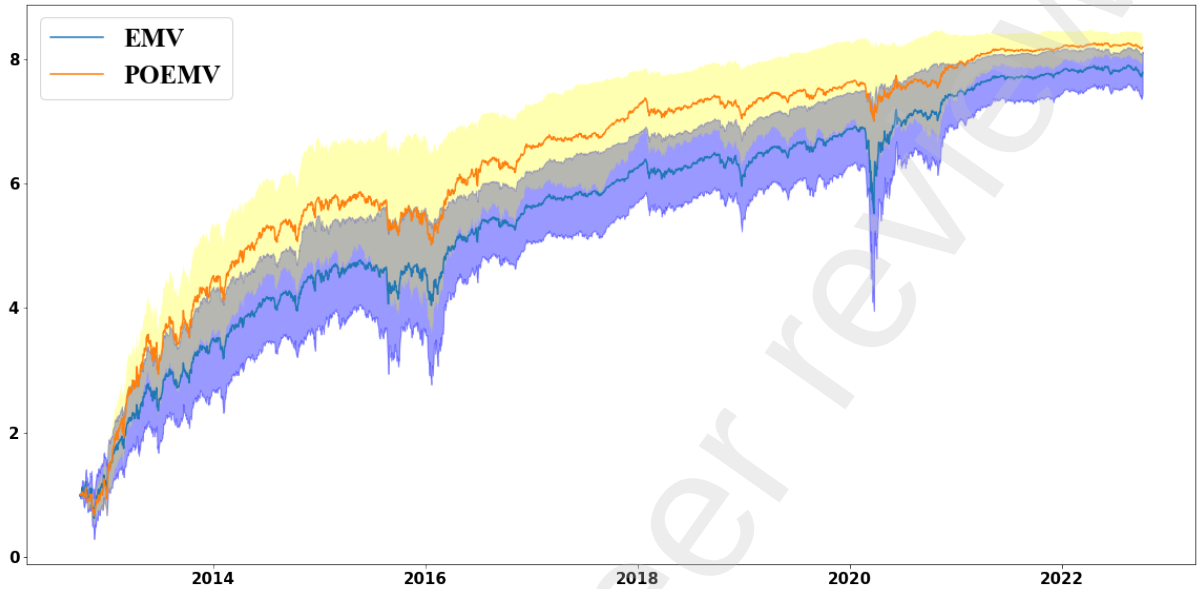
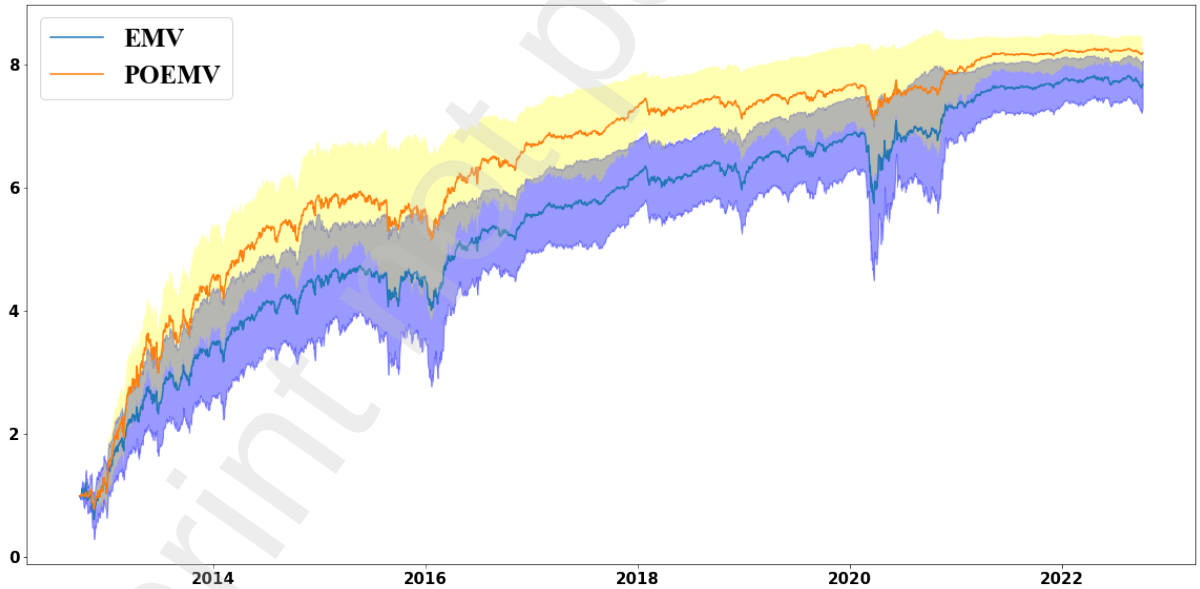
losses in the bear market). In particular, the POEMV policy shows much more resilience and stability during the stock market meltdown in March 2020.

We can also observe the effect of rebalancing frequency by comparing the daily and monthly results. Implementing the learned policy monthly can miss opportunities to adjust the investment policy timely as the market changes, and hence increases the variance of the terminal wealth for all three algorithms.[3]



(a) Daily rebalancing



(b) Monthly rebalancing

Figure 5: Wealth processes of EMV and POEMV over the 10-year out-of-sample period. The averaged wealth process plus/minus half standard deviation is plotted and the grey area shows their overlap.

[3]This comparison ignores transaction costs. Higher rebalancing frequency incurs more transaction costs, which would potentially affect the investment performance.

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	Mean	Variance	Sharpe ratio
POEMV	8.19	0.224	15.38
EMV	7.79	0.453	10.09
Classical	8.15	1.537	5.77

(a) Daily rebalancing

	Mean	Variance	Sharpe ratio
POEMV	8.19	0.284	13.49
EMV	7.69	0.597	8.66
Classical	8.22	10.081	2.27

(b) Monthly rebalancing

Table 4: Mean and variance of the terminal wealth and the 10-year Sharpe ratio of three algorithms

7 Conclusion

We have developed an actor-critic RL algorithm for continuous-time MV investment in a regime-switching market where the drift of the risky asset price is modulated by an unobservable Markov chain. Our algorithm is constructed based on the semi-analytical representations of the optimal value function and optimal policy of the exploratory MV problem with regime switches. Due to partial information, it requires filtering as a first step to obtain the belief probability of the market regime, which is then used as a feature to learn the optimal policy. Through simulation and empirical studies, we demonstrate its good empirical performance and significant advantages over the classical stochastic control approach and the EMV algorithm that learns the optimal MV policy without considering the market regime. In future research, it would be interesting to develop RL algorithms for investment and trading problems with regime switching under other types of criteria.

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A Multiple Risky Assets

There is no essential technical difficulty in extending our method to the multi-asset case. We consider the MV problem (1), where there are d risky assets, a risk-free asset that pays a constant interest $r \geq 0$, and the market has two regimes. We introduce some notations. By convention, all vectors are column vectors unless otherwise specified. The transpose and inverse of A are denoted by $A^\top$ and A^{-1}, respectively. For any vector v, $\|v\|$ is its Euclidean norm, and for a square matrix A, $|A|$ denotes its determinant.

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The price vector of d risky assets follows the SDE

$$dS_t = \mathrm{Diag}\left(S_t\right)\left(\mu_t dt + \sigma dB_t\right),$$

where $B := \{B_t : t \geq 0\}$ is a d-dimensional standard Brownian motion, and the volatility matrix $\sigma \in \mathbb{R}^{d\times d}$ is constant and non-degenerate. The drift vector μ_t is modulated by the CTMC I defined in Section 2.1. In regime i $(i = 1, 2)$, the stock mean return vector is given by $\mu^{(i)} \in \mathbb{R}^d$ and $\mu_t = \mu^{(i)}$ if $I_t = i$. We also define the Sharpe ratio vector $\rho^{(i)} := \sigma^{-1}(\mu^{(i)} - r \times \mathbb{1}_d) \in \mathbb{R}^d$, where $\mathbb{1}_d$ is a d-dimensional all-ones vector. We have $\rho_t = \rho^{(i)}$ if $I_t = i$.

Let $u_t \in \mathbb{R}^d$ be the vector of discounted dollar values in these d risky assets at time t. The discounted wealth follows

$$dX_t = u_t^\top \sigma\left(\rho_t dt + dB_t\right), \quad 0 \leq t \leq T.$$

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Let $\{\mathcal{F}_t^S : t \geq 0\}$ be the filtration generated by the d risky assets. The innovation process is given by

$$d\hat{B}_t = \sigma^{-1} d\ln(S_t) - \sigma^{-1}\left(\hat{\mu}_t - \frac{1}{2}\operatorname{Diag}(\sigma\sigma^\top)\right)dt,$$

where $\hat{\mu}_t = \mathbb{E}\left[\mu_t \mid \mathcal{F}_t^S\right] = p_t\mu^{(1)} + (1-p_t)\mu^{(2)}$ and $\operatorname{Diag}(\sigma\sigma^\top) = \left(a^{(1)}, \ldots, a^{(d)}\right)^\top$ with $a^{(i)} := \sum_{k=1}^d \sigma_{i,k}\sigma_{i,k}$. The innovation process $\hat{B} := \{\hat{B}_t : t \geq 0\}$ is a d-dimensional Brownian motion adapted to the observable filtration $\{\mathcal{F}_t^S : t \in [0,T]\}$.

We consider the belief probability of the market regime $p_t := P\left(I_t = 1 \mid \mathcal{F}_t^S\right)$, and its dynamics is given by the Wonham filter

$$dp_t = \left[-\left(\lambda_1 + \lambda_2\right)p_t + \lambda_2\right]dt + p_t\left(1-p_t\right)\left(\rho^{(1)} - \rho^{(2)}\right)^\top d\hat{B}_t$$

The dynamics of the discounted wealth can be rewritten as

$$dX_t^u = u_t^\top \sigma\left(\hat{\rho}_t dt + d\hat{B}_t\right), \quad 0 \leq t \leq T,$$

where $\hat{\rho}_t = p_t\rho^{(1)} + (1-p_t)\rho^{(2)}$.

To do RL, we consider stochastic policies of the feedback form $\boldsymbol{\pi} : (t,x,p) \in [0,T] \times \mathbb{R}^d \times (0,1) \mapsto \boldsymbol{\pi}(\cdot \mid t,x,p) \in \mathcal{P}(\mathbb{R}^d)$, where $\mathcal{P}(\mathbb{R}^d)$ denotes the class of probability density functions on $\mathbb{R}^d$. We also expand the filtered probability space $\left(\Omega, \mathcal{F}, \mathbb{P}; \{\mathcal{F}_t^S : t \in [0,T]\}\right)$ to $\left(\Omega, \mathcal{F}, \overline{\mathbb{P}}; \{\overline{\mathcal{F}}_t : t \in [0,T]\}\right)$ to accommodate action randomization in the same way as in Section 2.3. The discounted wealth process under stochastic policy $\boldsymbol{\pi}$ is given by

$$dX_t^{\boldsymbol{\pi}} = (u_t^{\boldsymbol{\pi}})^\top \sigma\left(\hat{\rho}_t dt + d\hat{B}_t\right), \quad 0 \leq t \leq T,$$

where $u_t^{\boldsymbol{\pi}}$ is sampled from $\boldsymbol{\pi}(\cdot \mid t, X_t^{\boldsymbol{\pi}}, p_t)$. The exploratory version of the discounted wealth follows

$$\begin{aligned} d\widetilde{X}_t = &\int_{\mathbb{R}^d} u^\top \sigma\hat{\rho}_t \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du dt + \int_{\mathbb{R}^d} \sigma u \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du d\hat{B}_t \\ &+ \sqrt{\int_{\mathbb{R}^d} u^\top \sigma\sigma^\top u \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du - \left\|\int_{\mathbb{R}^d} \sigma u \boldsymbol{\pi}\left(u \mid t, \widetilde{X}_t^{\boldsymbol{\pi}}, p_t\right) du\right\|^2} d\bar{B}_t, \end{aligned}$$

where $\bar{B}_t$ is another one-dimensional Brownian motion independent of $\hat{B}_t$, adapted to $\{\overline{\mathcal{F}}_t : t \in [0,T]\}$, and it is introduced due to the randomness caused by exploration.

The RSEMV problem in the general case still solves (13). Using similar arguments as in the case with one risky asset, we can derive that the optimal value function satisfies the HJB equation

$$\begin{aligned} &v_t + \min_{\boldsymbol{\pi}}\left\{\int_{\mathbb{R}^d}\left[\frac{1}{2}u^\top\sigma\sigma^\top u v_{xx} + u^\top\sigma\hat{\rho}v_x + u^\top\sigma(\rho^{(1)} - \rho^{(2)})p(1-p)v_{xp} + \Lambda\ln\boldsymbol{\pi}(u)\right]\boldsymbol{\pi}(u)du\right\} \\ &+ \frac{1}{2}\|\rho^{(1)} - \rho^{(2)}\|^2 p^2\left(1-p\right)^2 v_{pp} + \left[\lambda_2 - \left(\lambda_1 + \lambda_2\right)p\right]v_p = 0, \end{aligned}$$

with terminal condition $v(T,x,p) = (x-\omega)^2 - (\omega - z)^2$ and $\hat{\rho} = p\rho^{(1)} + (1-p)\rho^{(2)}$. Solving the optimization problem in the HJB equation, we obtain a feedback distribution, whose density is given by

$$\begin{aligned} \boldsymbol{\pi}^*(u \mid t,x,p) &= \frac{\exp\left(-\frac{1}{\Lambda}\left(\frac{1}{2}u^\top\sigma\sigma^\top u v_{xx} + u^\top\sigma[\hat{\rho}v_x + (\rho^{(1)} - \rho^{(2)})p(1-p)v_{xp}]\right)\right)}{\int_{\mathbb{R}^d}\exp\left(-\frac{1}{\Lambda}\left(\frac{1}{2}u^\top\sigma\sigma^\top u v_{xx} + u^\top\sigma[\hat{\rho}v_x + (\rho^{(1)} - \rho^{(2)})p(1-p)v_{xp}]\right)\right)du} \\ &= \mathcal{N}\left(u \mid -(\sigma^\top)^{-1}\frac{\hat{\rho}v_x + (\rho^{(1)} - \rho^{(2)})p(1-p)v_{xp}}{v_{xx}}, \left(\sigma\sigma^\top\right)^{-1}\frac{\Lambda}{v_{xx}}\right), \end{aligned}$$

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which is a multivariate Gaussian density. Substituting it back to the HJB equation, we would obtain a nonlinear PDE

$$v_t - \frac{\|\hat{\rho} v_x + \left(\rho^{(1)} - \rho^{(2)}\right) p\,(1-p)\, v_{xp}\|^2}{2 v_{xx}} - \frac{\Lambda d}{2} \ln\left(\frac{2\pi\Lambda}{|\sigma\sigma^\top|\, v_{xx}}\right)$$
$$+ \frac{1}{2}\|\rho^{(1)} - \rho^{(2)}\|^2 p^2 (1-p)^2 v_{pp} + \left[\lambda_2 - (\lambda_1 + \lambda_2)\, p\right] v_p = 0.$$

We still use the ansatz (16) to solve this PDE. After some calculations, we obtain

$$f_t + \frac{1}{2}\|\rho^{(1)} - \rho^{(2)}\|^2 p^2 (1-p)^2 f_{pp} - \left[2\hat{\rho}^\top \left(\rho^{(1)} - \rho^{(2)}\right) p\,(1-p) + (\lambda_1 + \lambda_2)\, p - \lambda_2\right] f_p$$
$$+ \|\hat{\rho}\|^2 f = 0,$$
$$g_t + \frac{1}{2}\|\rho^{(1)} - \rho^{(2)}\|^2 p^2 (1-p)^2 g_{pp} + \left[\lambda_2 - (\lambda_1 + \lambda_2)\, p\right] g_p - \frac{\Lambda d}{2} \ln\left(\frac{\pi\Lambda f}{|\sigma\sigma^\top|}\right) = 0,$$

for $(t,p) \in [0,T) \times (0,1)$. The terminal conditions are $f(T,p) = 1$ and $g(T,p) = 0$ for $p \in (0,1)$. The two functions $f(t,p)$ and $g(t,p)$ still satisfy the properties listed in Theorem 1. Thus, we can use similar arguments to establish a verification theorem and show that the ansatz still gives the optimal value function and the optimal feedback control distribution is given by

$$\mathcal{N}\left(u \mid -(x-\omega)(\sigma^\top)^{-1}\left(\hat{\rho} - \left(\rho^{(1)} - \rho^{(2)}\right) p(1-p)\frac{f_p(t,p)}{f(t,p)}\right), \left(\sigma\sigma^\top\right)^{-1} \frac{f(t,p)\Lambda}{2}\right),$$

We still parametrize $f(t,p)$ and $g(t,p)$ by polynomials of order m as in (25) and (26), respectively, and parametrize the value function as in (24). We parametrize the policy as

$$\boldsymbol{\pi}^\phi\left(u \mid t,x,p\right) = \mathcal{N}\left(u \mid -(x-\omega)\left(\phi_1\left(p - \frac{f_p^\theta(t,p)}{f^\theta(t,p)} p(1-p)\right) + \phi_2\right), \Sigma \frac{f^\theta(t,p)\Lambda}{2}\right),$$

and denote its mean vector by μ^ϕ and covariance matrix by σ^ϕ. We put $\phi = (\phi_1, \phi_2, \Sigma) \in \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{S}^d_{++}$, where $\mathbb{S}^d_{++}$ is the set of $d \times d$ symmetric and positive definite matrices. As $\boldsymbol{\pi}^\phi$ is a d-dimensional Gaussian density, its differential entropy is given by

$$\mathcal{H}(\boldsymbol{\pi}^\phi\left(\cdot \mid t,x,p\right)) = \frac{d}{2}\ln(2\pi e) + \frac{1}{2}\ln\left|\sigma^\phi\right|,$$

which does not depend on x.

We can perform PE and PG updates in the same way the one risky asset case.

PE update

$$\theta \leftarrow \theta + \eta_\theta \left\{\sum_{k=0}^{K-1}\left[\left(h\left(X_T^{\boldsymbol{\pi}^\phi}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right) - \sum_{i=k}^{K-1} \Lambda \mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_i, X_{t_i}^{\boldsymbol{\pi}^\phi}, p_{t_i}\right)\right)\Delta t\right)\right.\right.$$
$$\left.\left. \times \frac{\partial V^\theta}{\partial \theta}(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k})\right]\Delta t\right\},$$

where $\frac{\partial V^\theta}{\partial \theta_{i,j}}(t,x,p)$ and $\frac{\partial V^\theta}{\partial \tilde{\theta}_{i,j}}(t,x,p)$ are still given by (32) and (33), respectively.

PG update

$$\phi \leftarrow \phi - \eta_\phi \left\{\sum_{k=0}^{K-1} \frac{\partial}{\partial \phi} \ln \boldsymbol{\pi}^\phi\left(u_{t_k} | t_k, X_{t_k}^{\boldsymbol{\pi}_k^\phi}, p_{t_k}\right)\left[V^\theta\left(t_{k+1}, X_{t_{k+1}}^{\boldsymbol{\pi}^\phi}, p_{t_{k+1}}\right) - V^\theta\left(t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right.\right.$$
$$\left.\left. - \Lambda \mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right)\Delta t\right] - \Lambda \frac{\partial}{\partial \phi}\mathcal{H}\left(\boldsymbol{\pi}^\phi\left(\cdot \mid t_k, X_{t_k}^{\boldsymbol{\pi}^\phi}, p_{t_k}\right)\right)\Delta t\right\},$$

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where

$$\frac{\partial}{\partial \phi_1} \ln \boldsymbol{\pi}^{\phi}(u \mid t,x,p) = -(x-\omega)\left(\sigma^{\phi}\right)^{-1}\left(u-\mu^{\phi}\right)\left[p - \frac{f_p^{\theta}(t,p)}{f^{\theta}(t,p)}p(1-p)\right],$$

$$\frac{\partial}{\partial \phi_2} \ln \boldsymbol{\pi}^{\phi}(u \mid t,x,p) = -(x-\omega)\left(\sigma^{\phi}\right)^{-1}\left(u-\mu^{\phi}\right),$$

$$\frac{\partial}{\partial \Sigma} \ln \boldsymbol{\pi}^{\phi}(u \mid t,x,p) = \frac{1}{2}(\Sigma)^{-1}\left[\left(u-\mu^{\phi}\right)\left(u-\mu^{\phi}\right)^{\top}\left(\sigma^{\phi}\right)^{-1} - I_d\right],$$

$$\frac{\partial}{\partial \phi_1}\mathcal{H}\left(\boldsymbol{\pi}^{\phi}(\cdot \mid t,x,p)\right) = \frac{\partial}{\partial \phi_2}\mathcal{H}\left(\boldsymbol{\pi}^{\phi}(\cdot \mid t,x,p)\right) = 0,$$

$$\frac{\partial}{\partial \Sigma}\mathcal{H}\left(\boldsymbol{\pi}^{\phi}(\cdot \mid t,x,p)\right) = \frac{1}{2}(\Sigma)^{-1},$$

where I_d is the $d \times d$ identity matrix.

B Finite Difference Scheme

We present a finite difference scheme to solve PDEs (17) and (18). Both of them can be cast into the form

$$u_t + b(p)u_{pp} + a(p)u_p + c(p)u = h(t,p),\ \ p \in (0,1), t \in [0,T), \quad u(T,p) = \phi(p).$$

We create a uniform grid $\{(t_i, p_j) : i = 0, \cdots, M, j = 0, \cdots, N\}$ in the $[0,T] \times [0,1]$ space with Δt and Δp as the step size for t and p, respectively. We set $t_0 = 0$, $t_M = T$, $p_0 = 0$ and $p_M = 1$. We use a uniform grid here for the simplicity of presentation, but non-uniform grids can also be adopted. In the following, we simply write u_j^i for $u(t_i, p_j)$ and h_j^i for $h(t_i, p_j)$. We numerically solve the PDE by the popular Crank-Nicolson scheme; see e.g., Duffy (2013) for an introduction. For u_p, we approximate it by the upwind finite difference scheme because $a(p)$ can change signs, i.e.,

$$u_p(t_i,p_j) \approx \begin{cases} \frac{(u_{j+1}^i - u_j^i) + (u_{j+1}^{i+1} - u_j^{i+1})}{2\Delta p}\mathbb{1}_{\{a(p)\geq 0\}}, \\ \frac{(u_j^i - u_{j-1}^i) + (u_j^{i+1} - u_{j-1}^{i+1})}{2\Delta p}\mathbb{1}_{\{a(p)<0\}}. \end{cases}$$

This can get rid of spurious oscillations of the solution near the boundary in our problem. For the other terms, we approximate them as

$$u_t(t_i,p_j) \approx \frac{u_j^{i+1} - u_j^i}{\Delta t},$$

$$u_{pp}(t_i,p_j) \approx \frac{u_{j+1}^i - 2u_j^i + u_{j-1}^i + u_{j+1}^{i+1} - 2u_j^{i+1} + u_{j-1}^{i+1}}{2(\Delta p)^2},$$

$$u(t_i,p_j) \approx \frac{u_j^{i+1} + u_j^i}{2}.$$

Using these approximations, we obtain

$$\begin{aligned} & u_{j-1}^i\left(\frac{-a(p)}{2\Delta p}\mathbb{1}\{a(p)<0\} + \frac{b(p)}{2(\Delta p)^2}\right) + u_j^i\left(-\frac{1}{\Delta t} - \frac{b(p)}{(\Delta p)^2} + \frac{c(p)}{2} - \frac{a(p)}{2\Delta p}\operatorname{sgn}(a(p))\right) \\ & + u_{j+1}^i\left(\frac{a(p)}{2\Delta p}\mathbb{1}\{a(p)\geq 0\} + \frac{b(p)}{2(\Delta p)^2}\right) = -u_{j-1}^{i+1}\left(\frac{-a(p)}{2\Delta p}\mathbb{1}\{a(p)<0\} + \frac{b(p)}{2(\Delta p)^2}\right) \\ & - u_j^{i+1}\left(\frac{1}{\Delta t} - \frac{b(p)}{(\Delta p)^2} + \frac{c(p)}{2} - \frac{a(p)}{2\Delta p}\operatorname{sgn}(a(p))\right) - u_{j+1}^{i+1}\left(\frac{a(p)}{2\Delta p}\mathbb{1}\{a(p)\geq 0\} + \frac{b(p)}{2(\Delta p)^2}\right) \\ & + \frac{h_j^{i+1} + h_j^i}{2}. \end{aligned} \tag{38}$$

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The Crank-Nicolson scheme works recursively as follows. Set $u_j^M = \phi(p_j)$ from the terminal condition. Given $(u_0^{i+1}, \cdots, u_N^{i+1})$, we solve for $(u_0^i, \cdots, u_N^i)$ from the following linear system

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ A_1 & B_1 & C_1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & A_2 & B_2 & C_2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & A_3 & B_3 & C_3 & \cdots & 0 & 0 & 0 \\ \vdots & & & & & & & & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & C_{N-3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & B_{N-2} & C_{N-2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & A_{N-1} & B_{N-1} & C_{N-1} \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_0^i \\ u_1^i \\ u_2^i \\ u_3^i \\ \vdots \\ u_{N-3}^i \\ u_{N-2}^i \\ u_{N-1}^i \\ u_N^i \end{pmatrix} = \begin{pmatrix} D_0 \\ D_1 \\ D_2 \\ D_3 \\ \vdots \\ D_{N-3} \\ D_{N-2} \\ D_{N-1} \\ D_N \end{pmatrix}$$

where for $j = 1, \cdots, N-1$, A_j, B_j, and C_j are the coefficients of u_{j-1}^i, u_j^i, and u_{j+1}^i, respectively, and D_j is the sum on the right-hand-side of (38). The terms D_0 and D_N represent the boundary conditions at $p = 0$ and $p = 1$. Recall that the belief state process in our problem can never hit 0 or 1 when starting from the interior, and hence no boundary conditions are needed for the well-posedness of the PDEs (17) and (18). However, we need to impose artificial boundary conditions for the implementation of the numerical scheme. A natural choice is using the solutions in Wang and Zhou (2020) because $p = 0$ or 1 corresponds to the market having only one regime, which is the case studied in their paper. Specifically, from Wang and Zhou (2020), we can set

$$f(t,0) = e^{\rho_2^2(T-t)}, \;\; f(t,1) = e^{\rho_1^2(T-t)}, \quad t \in [0,T].$$

and

$$g(t,0) = \frac{\Lambda\rho_2^2}{4}\left(T^2 - t^2\right) - \frac{\Lambda}{2}\left(\rho_2^2 T - \ln\frac{\sigma^2}{\pi\Lambda}\right)(T-t), \quad t \in [0,T]$$
$$g(t,1) = \frac{\Lambda\rho_1^2}{4}\left(T^2 - t^2\right) - \frac{\Lambda}{2}\left(\rho_1^2 T - \ln\frac{\sigma^2}{\pi\Lambda}\right)(T-t), \quad t \in [0,T]$$

We then set D_0 and D_N at each time step t_i by using these formulas with $t = t_i$. A potential problem is that ρ_1 can be too large so that numerically calculating $e^{\rho_1^2(T-t)}$ is problematic. In this case, we can simply set $f(t,1)$ equal to some constant.

We first apply the above numerical scheme to solve for $f(t,p)$. We then apply it to solve for $g(t,p)$ by using the same grid as for $f(t,p)$ and the solution of $f(t,p)$ on the grid as $f(t,p)$ appears in the source term in PDE (18). In our implementation, we set $\Delta t = \frac{1}{252}$ and $\Delta p = \frac{1}{10000}$ to obtain accurate results.

C Proofs

Proof of Theorem 1: We consider the following domains:

$$\Omega = (0,T) \times (0,1), \;\; \Omega_0 = \Omega \cup \{\{0\} \times (0,1)\}, \;\; \partial_0\Omega = \{\{T\} \times (0,1)\}.$$

Let $\overline{\Omega}$ be the closure of Ω. The differential operator in both PDEs has the following form:

$$\mathcal{L}u := u_t + \frac{1}{2}a(p)u_{pp} + b(p)u_p + c(p)u,$$

and the PDEs can be cast in the form

$$\mathcal{L}u(t,p) = h(t,p), \;\; u(T,p) = \phi(p). \tag{39}$$

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For both PDEs, $a(p)$, $b(p)$, $c(p)$ and their derivatives are uniformly Lipschitz continuous and bounded on $[0,1]$. Moreover, $a(p) > 0$ for $p \in (0,1)$ and $a(p) = 0$ for $p = 0$ or 1.

Existence. We establish the existence of solutions to these two PDEs. Friedman (1976) constructed the fundamental solution to (39) for the degenerate case. We denote it by $\Gamma(t,p;\tau,\xi)$, which is in $C^{1,2}([0,T) \times [\varepsilon, 1-\varepsilon]) \cap C([0,T] \times [\varepsilon, 1-\varepsilon])$ for any $\varepsilon \in (0,1/2)$ as a function of (t,p) (see Theorem 1.2 in Chapter 15 of Friedman (1976)). A solution to (39) is given by

$$u(t,p) = \int_0^1 \Gamma(t,p;T,\xi)\phi(\xi)d\xi - \int_0^t \int_0^1 \Gamma(t,p;\tau,\xi)h(\tau,\xi)d\xi d\tau, \tag{40}$$

For PDE (17), $\phi(\xi) = 1$, $h(\tau,\xi) = 0$. For PDE (18), $\phi(\xi) = 0$, $h(\tau,\xi) = \frac{\Lambda}{2}\ln\frac{\pi\Lambda f(\tau,\xi)}{\sigma^2}$. The property that $u(t,p) \in C^{1,2}([0,T) \times [\varepsilon, 1-\varepsilon]) \cap C([0,T] \times [\varepsilon, 1-\varepsilon])$ for both PDEs follows from the representaion (40).

Below we will use the weak maximum (minimum) principle (see e.g., Lieberman (1996)) repeatedly for further analysis. We simply refer to it as the maximum (minimum) principle.

Uniqueness. Suppose there exist two solutions to (17), f_1 and f_2, and without loss of generality, we let $f_1 < f_2$ for some point (t_0, p_0) in Ω_0. However, $u = f_1 - f_2$ satisfies $\mathcal{L}u \le 0$ on Ω_0 and $u \ge 0$ on $\partial_0\Omega$. Applying the minimum principle, we deduce that $u = f_1 - f_2 \ge 0$ on Ω_0, which contradicts our initial assumption. Thus, the solution to (17) is unique. We can prove the uniqueness of the solution to (18) in the same way.

Boundedness. (i) We prove that the solution $f(t,p)$ to (17) is bounded over $[0,T] \times (0,1)$ by establishing its supersolution $\overline{f}(t,p)$ and subsolution $\underline{f}(t,p)$. Consider

$$\overline{\mathcal{L}}u := \mathcal{L}u + (\overline{\rho}^2 - \hat{\rho}^2)u \quad \text{and} \quad \underline{\mathcal{L}}u := \mathcal{L}u - \hat{\rho}^2 u,$$

where $\overline{\rho} = \max\{|\rho_1|, |\rho_2|\}$ and we have $\overline{\rho}^2 - \hat{\rho}^2 \ge 0$. We consider two PDEs

$$\begin{aligned} \overline{\mathcal{L}}u &= 0, \qquad u(t,p) = 1, \\ \underline{\mathcal{L}}u &= 0, \qquad u(t,p) = 1, \end{aligned}$$

The solution to the first one is given by $\overline{f} = e^{\overline{\rho}^2(T-t)}$ and the solution to the second one is given by $\underline{f} = 1$. We show that $\underline{f}(t,p) \le f(t,p) \le \overline{f}(t,p)$ for $(t,p) \in [0,T] \times (0,1)$ in the following.

As $c(p) = \hat{\rho}^2 \ge 0$, $\mathcal{L}(\overline{f} - \underline{f}) \le 0$ on Ω_0 and $f - \underline{f} = 0$ on $\partial\Omega_0$, we obtain $f(t,p) - \underline{f}(t,p) \ge 0$ using the minimum principle.

Let $u = e^{-\overline{\rho}^2(T-t)}f$. Some calculations yield the PDE for u given by

$$\tilde{\mathcal{L}}u = 0, \text{ where } \tilde{\mathcal{L}}u = u_t + \frac{1}{2}a(p)u_{pp} + b(p)u_p + \tilde{c}(p)u, \ \tilde{c}(p) = \hat{\rho}^2 - \overline{\rho}^2.$$

As $\tilde{c}(p) \le 0$ and $\tilde{\mathcal{L}}u \ge 0$, we infer $u \le 1$ from the maximum principle. Thus, we have $f(t,p) \le e^{\overline{\rho}^2(T-t)} = \overline{f}(t,p)$. The lower bound of f implies that $v_{xx}(t,x,p) = 2/f > 0$, where $v(t,x,p)$ is given by (16).

(ii) We prove that $f_p(t,p)$ is bounded over $[0,T] \times (0,1)$. Differentiating both sides of (17) with respect to p, we obtain

$$\partial_t(f_p) + a(p)\partial_{pp}(f_p) + b_1(p)\partial_p(f_p) + c_1(p)(f_p) + d_1(p)f = 0,$$

with terminal condition $f_p(T,p) = 0$. Note that $a(p)$ is the same as in (17). The new coefficients $b_1(p), c_1(p), d_1(p)$ are polynomials of p. Thus, they are uniformly Lipschitz continuous and bounded on $[0,1]$. We also have $|f(t,p)| \le e^{\overline{\rho}^2 T}$. Hence, there exists a sufficiently large constant $L > 0$ such that $|c_1(p)| \le L$ and $|d_1(p)f| \le L$. We consider $u(t,p) = e^{-L(T-t)}f_p(t,p)$ and obtain

$$\tilde{\mathcal{L}}'u = -e^{-L(T-t)}d_1(p)f, \text{ where } \tilde{\mathcal{L}}'u = u_t + \frac{1}{2}a(p)u_{pp} + b_1(p)u_p + \tilde{c}_1(p)u, \ \tilde{c}_1(p) = c_1(p) - L.$$

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As $\tilde{c}_1(p) \leq 0$ and $\tilde{\mathcal{L}}'(u - L(T-t)) = -e^{-L(T-t)}d_1(p)f + L - \tilde{c}_1(p)L(T-t) \geq 0$, we obtain $u - L(T-t) \leq 0$ by applying the maximum principle. As a result, $f_p \leq L(T-t)e^{L(T-t)} \leq K$, where $K = LTe^{LT}$, which is entirely determined by the environment parameters. We can also establish $f_p \geq -K$ by considering $u = -f_p$ and obtaining the upper bound of u in the same way. Together, we have shown that $|f_p(t,p)| \leq K$.

(iii) We derive the upper and lower bounds of $g(t,p)$, the solution to (18). Recall that

$$\mathcal{L}g = h, \text{ where } \mathcal{L}g = g_t + \frac{1}{2}a(p)g_{pp} + b(p)g_p, \ h = \frac{\Lambda}{2}\ln\frac{\pi\Lambda f}{\sigma^2}.$$

Using the bounds for f, we have

$$C_1 \leq h \leq C_2, \text{ where } C_1 = \frac{\Lambda}{2}\ln\frac{\pi\Lambda}{\sigma^2}, \ C_2 = \frac{\Lambda}{2}(\ln\frac{\pi\Lambda}{\sigma^2} + \bar{\rho}^2 T).$$

It follows that

$$\mathcal{L}(g + C_1(T-t)) \geq 0 \quad \text{and} \quad \mathcal{L}(g + C_2(T-t)) \leq 0.$$

Note that $g = 0$ on $\partial_0\Omega$. For the first inequality, using the maximum principle, we obtain $g + C_1(T-t) \leq 0$, which implies $g \leq -C_1(T-t)$. Similarly, using the minimum principle for the second inequality, we obtain $g \geq -C_2(T-t)$. Setting $c = -C_2$ and $C = -C_1$ gives us the desired claim.

Monotonicity of $f(t,p)$ as a function of t. Differentiating both sides of (17) with respect to t, we obtain

$$\partial_t(f_t) + a(p)\partial_{pp}(f_t) + b(p)\partial_p(f_t) + c(p)(f_t) = 0,$$

with terminal condition $f_t(T,p) \leq 0$, which holds because $1 \leq f(t,p) \leq e^{\bar{\rho}^2(T-t)}$ and $f(T,p) = 1$. We can show $f_t \leq 0$ by applying the maximum principle to $u(t,p) = e^{-\bar{\rho}^2(T-t)}f_t(t,p)$. Thus, for any $p \in (0,1)$, $f(t,p)$ is a decreasing function of t. □

Proof of Theorem 2: We first show that the feedback policy $\boldsymbol{\pi}^*$ given by (20) is admissible. Condition (i) of admissibility is clearly verified by $\boldsymbol{\pi}^*$. For condition (ii) of admissibility, the exploratory discounted wealth process under $\boldsymbol{\pi}^*$ follows

$$\begin{aligned} d\widetilde{X}_t^* =& -\hat{\rho}_t(\widetilde{X}_t^* - \omega)\left[\hat{\rho}_t - (\rho_1 - \rho_2)\, p_t(1-p_t)\frac{f_p(t,p_t)}{f(t,p_t)}\right]dt \\ & - (\widetilde{X}_t^* - \omega)\left[\hat{\rho}_t - (\rho_1 - \rho_2)\, p_t(1-p_t)\frac{f_p(t,p_t)}{f(t,p_t)}\right]d\hat{B}_t + \sqrt{\frac{f(t,p_t)\Lambda}{2}}d\bar{B}_t, \end{aligned} \tag{41}$$

with the initial state $\widetilde{X}_0^* = x$. The drift and volatility coefficients of the SDE (41) are linear in X, and $f(t,p)$ and $f_p(t,p)$ are bounded for $(t,p) \in [0,T] \times (0,1)$ from Theorem 1. Therefore, these coefficients satisfy linear growth and global Lipschitz conditions, which guarantees a unique strong solution to (41), and we also have

$$\mathbb{E}\left[\sup_{0\leqslant t\leqslant T}\left|\widetilde{X}_t^*\right|^2\right] \leq C\left(1 + x^2\right)e^{CT}, \tag{42}$$

for some constant $C > 0$; see e.g., Karatzas and Shreve (1991). Thus, condition (ii) of admissibility is verified. For condition (iii) of admissibility, we have

$$\begin{aligned} &\mathbb{E}\left[\int_0^T\int_{\mathbb{R}} u^2\boldsymbol{\pi}^*\left(u \mid s, \widetilde{X}_s^*, p_s\right)duds\right] \\ =&\mathbb{E}\left[\int_0^T\left(\frac{f(t,p_t)\Lambda}{2\sigma^2} + \frac{(\widetilde{X}_t^* - \omega)^2}{\sigma^2}\left(\rho_2 + (\rho_1 - \rho_2)p_t - (\rho_1 - \rho_2)\, p_t(1-p_t)\frac{f_p(t,p_t)}{f(t,p_t)}\right)^2\right)dt\right] \\ \leq& C_1 + C_2\mathbb{E}\left[\int_0^T(\widetilde{X}_t^* - \omega)^2dt\right] \end{aligned}$$

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for some constant $C_1, C_2 > 0$, where we use Theorem 1 to bound the terms involving p_t. The validity of condition (iii) then follows from (42). For condition (iv) of admissibility, the first term is clearly finite due to (42). For the second term, we have

$$\begin{aligned}
&\mathbb{E}\left[\left|\Lambda \int_0^T \int_{\mathbb{R}} \boldsymbol{\pi}^*\left(u \mid s, \widetilde{X}_s^*, p_s\right) \ln \boldsymbol{\pi}^*\left(u \mid s, \widetilde{X}_s^*, p_s\right) du ds\right|\right] \\
\leq & \Lambda \int_0^T \mathbb{E}\left[\left|\int_{\mathbb{R}} \boldsymbol{\pi}^*\left(u \mid s, \widetilde{X}_s^*, p_s\right) \ln \boldsymbol{\pi}^*\left(u \mid s, \widetilde{X}_s^*, p_s\right) du\right|\right] ds \\
= & \frac{\Lambda}{2} \int_0^T \mathbb{E}\left[\left|\ln \frac{\pi e \Lambda}{\sigma^2} + \ln f(s, p_s)\right|\right] ds,
\end{aligned}$$

which is also finite due to the boundedness of f. Thus, $\boldsymbol{\pi}^*$ is admissible.

Now consider an arbitrary admissible feedback policy $\boldsymbol{\pi}$. We introduce the infinitesimal generator of the process $(\widetilde{X}_t^{\boldsymbol{\pi}}, p_t)$, whose action on a sufficiently smooth function $w(t, x, p)$ is given by

$$\begin{aligned}
\mathcal{G}^{\boldsymbol{\pi}} w(t, x, p) := & \int_{\mathbb{R}} \left[u \sigma \hat{\rho} w_x + u \sigma \left(\rho_1 - \rho_2\right) p(1-p) w_{xp} + \frac{1}{2} \sigma^2 u^2 w_{xx}\right] \boldsymbol{\pi}\left(u \mid t, x, p\right) du \\
& + \left[-\left(\lambda_1 + \lambda_2\right) p + \lambda_2\right] w_p + \frac{1}{2}\left(\rho_1 - \rho_2\right)^2 p^2 (1-p)^2 w_{pp}.
\end{aligned}$$

Consider the following stopping times for any fixed t:

$$\tau_n := \inf\{s \geq t : p_s \in (0, 1/n] \cup [1 - 1/n, 1) \text{ or } |\widetilde{X}_s^{\boldsymbol{\pi}}| \geq n\}, \quad n = 1, 2 \cdots.$$

As 0 and 1 are entrance boundaries, $\tau_n \nearrow \infty$ as n goes to infinity.

For $v(t, x, p)$ given by (16), we apply Ito's formula to $v(t \wedge \tau_n, \widetilde{X}_{t \wedge \tau_n}^{\boldsymbol{\pi}}, p_{t \wedge \tau_n})$ and obtain

$$\begin{aligned}
& v\left(T \wedge \tau_n, \widetilde{X}_{T \wedge \tau_n}^{\boldsymbol{\pi}}, p_{T \wedge \tau_n}\right) \\
= & v(t, x, p) + \int_t^{T \wedge \tau_n} \left[v_s(s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s) + \mathcal{G}^{\boldsymbol{\pi}} v(s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s)\right] ds \\
& + \int_t^{T \wedge \tau_n} \frac{2(\widetilde{X}_s^{\boldsymbol{\pi}} - \omega)\sigma}{f(s, p_s)} \int_{\mathbb{R}} u \boldsymbol{\pi}\left(u \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s\right) du d\hat{B}_s \\
& + \int_t^{T \wedge \tau_n} \left(g_p(s, p_s) - (\widetilde{X}_s^{\boldsymbol{\pi}} - \omega)^2 \frac{f_p(s, p_s)}{f^2(s, p_s)}\right) \left(\rho_1 - \rho_2\right) p_s (1 - p_s) d\hat{B}_s \\
& + \int_t^{T \wedge \tau_n} \frac{2(\widetilde{X}_s^{\boldsymbol{\pi}} - \omega)\sigma}{f(s, p_s, s)} \sqrt{\int_{\mathbb{R}} u^2 \boldsymbol{\pi}\left(u \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s\right) du - \left(\int_{\mathbb{R}} u \boldsymbol{\pi}\left(u \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s\right) du\right)^2} d\bar{B}_s,
\end{aligned}$$

The three stochastic integrals driven by Brownian motions are martingales because the integrand processes are square integrable, as seen from the boundedness of $\widetilde{X}_s^{\boldsymbol{\pi}}, f, f_p, g, g_p$ up to $T \wedge \tau_n$ and condition (iii) of admissibility satisfied by $\boldsymbol{\pi}$. Thus, after taking expectations on both sides, we obtain

$$\begin{aligned}
& \mathbb{E}\left[v\left(T \wedge \tau_n, \widetilde{X}_{T \wedge \tau_n}^{\boldsymbol{\pi}}, p_{T \wedge \tau_n}\right)\right] \\
& = v(t, x, p) + \mathbb{E}_{t,x,p}\left[\int_t^{T \wedge \tau_n} v_s(s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s) + \mathcal{G}^{\boldsymbol{\pi}} v(s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s) ds\right] \\
& \geq v(t, x, p) - \Lambda \int_t^{T \wedge \tau_n} \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s)\right) ds,
\end{aligned}$$

where we use the HJB equation to obtain the inequality. It follows that

$$v(t, x, p) \leq \mathbb{E}_{t,x,p}\left[\left(\widetilde{X}_{T \wedge \tau_n}^{\boldsymbol{\pi}} - \omega\right)^2 - \Lambda \int_t^{T \wedge \tau_n} \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s)\right) ds\right] - (\omega - z)^2,$$

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As $\boldsymbol{\pi}$ is admissible, condition (iv) of admissibility allows us to use the dominated convergence theorem. Sending n to ∞, we obtain

$$v(t,x,p) \leq \mathbb{E}_{t,x,p}\left[\left(\widetilde{X}_T^{\boldsymbol{\pi}} - \omega\right)^2 - \Lambda \int_t^T \mathcal{H}\left(\boldsymbol{\pi}(\cdot \mid s, \widetilde{X}_s^{\boldsymbol{\pi}}, p_s)\right) ds\right] - (\omega - z)^2, \tag{43}$$

As $\boldsymbol{\pi}$ is chosen arbitrarily, we obtain $v(t,x,p) \leq V^*(t,x,p)$ for all $(t,x,p) \in [0,T] \times \mathbb{R} \times (0,1)$. If $\boldsymbol{\pi} = \boldsymbol{\pi}^*$, we see that equality holds in (43). Therefore, we conclude $v(t,x,p)$ is the optimal value function and $\boldsymbol{\pi}^*$ is the optimal feedback policy. □

Proof of Corollary 1: For the classical optimal control problem (2) without exploration, we can solve it in the same way as we solve the exploratory problem by considering the ansatz (16) with $g(t,p) \equiv 0$. Using the properties of $f(t,p)$ established in Theorem 2, we can verify that its optimal value function is given by (22) with the optimal feedback control given by $m^*(t,x,p)$.

To show convergence of the exploratory solution to the classical solution, note that

$$\lim_{\Lambda \to 0} \frac{\Lambda}{2} \ln \frac{\pi \Lambda f}{\sigma^2} = 0.$$

As the source term in the PDE (18) vanishes, $g(t,p;\Lambda)$ converges to the zero function. Hence, we obtain convergence of the exploratory solution to the classical solution.

For the optimal control distribution given by (20), its variance vanishes as $\Lambda \to 0$. Thus, we have the weak convergence to the Dirac measure concentrated at $m^*(t,x,p)$. □

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